Mineralium Deposita
https://doi.org/10.1007/s00126-025-01392-6

ARTICLE

Trace element zoning and temperature variation in sphalerite from the giant Nannihu porphyry-skarn-epithermal polymetallic system, central China

Chang Wang[1] · Zhan-Ke Li[1,2] · Ioan Sanislav[3] · Yi Zheng[4] · Xiao-Ming Li[1] · Rong-Zhen Zhang[5] · Jia Hu[1] · Jun Zhang[1]

Received: 29 August 2024 / Accepted: 21 August 2025

Abstract

The Nannihu ore field at the southern margin of the North China Craton is a typical magmatic-hydrothermal metallogenic system consisting of the large Nannihu porphyry Mo-W deposit, Luotuoshan skarn Zn-Cu-W deposit, and Lengshuibeigou epithermal Pb-Zn-Ag deposit. Sphalerite is present as a minor to major phase in all these deposits, providing an ideal object to investigate its trace element zoning and temperature variation. The field data and associated mineral assemblages indicate that only one type of sphalerite is present at the Nannihu porphyry deposit, whereas two types of sphalerite are present at both the Luotuoshan skarn deposit and Lengshuibeigou epithermal deposit. Sphalerite from the Nannihu porphyry deposit has Ga, Cd and In concentrations of 0.64–7.26 ppm, 1426–1687 ppm and 291–460 ppm, respectively. In the Luotuoshan skarn deposit, sphalerite contains relatively high In (147–616 ppm) and moderate Cd (1234–1558 ppm), with low Ga (below 43 ppm). Sphalerite from the Lengshuibeigou epithermal deposit has Ga, Cd, and In contents of 3.70–589 ppm, 1453–2673 ppm, and 26.7–308 ppm, respectively. Germanium concentrations are low in sphalerite (<5 ppm) from all three deposits. Infrared microthermometry of sphalerite-hosted fluid inclusions reveals that the sphalerite in the Luotuoshan deposit precipitated from intermediate- to low-salinity fluids (4.0 ~ 13.7 wt% NaCl equiv) at 398 to 308 °C, whereas the Lengshuibeigou deposit formed from low-salinity fluids (2.5 ~ 7.4 wt% NaCl equiv) at 305 to 238 °C. Indium tends to be enriched in sphalerite of the proximal high-temperature Luotuoshan skarn deposit, whereas Ga is more concentrated in sphalerite from the distal lower-temperature Lengshuibeigou deposit. Gallium and In distributions, combined with data collected from other magmatic-hydrothermal ore fields and deposits, reveal that In-enriched sphalerite occurs in proximal skarn deposits of magmatic-hydrothermal systems, and Ga-enriched sphalerite usually occurs in distal and shallow epithermal deposits. Ga/In values of sphalerite are related to the distance between mineralization and the causative intrusion of the magmatic-hydrothermal systems, and can be an indicator for exploration targeting.

Keywords Sphalerite · Trace elements · Indium · Gallium · Nannihu · Magmatic-hydrothermal system

Editorial handling: D. Zhai

✉ Zhan-Ke Li
lizk@cug.edu.cn

1 School of Earth Resources, China University of Geosciences, Wuhan 430074, Hubei Province, China

2 State Key Laboratory of Geological Processes and Mineral Resources, University of Geosciences, Wuhan 430074, Hubei Province, China

3 Economic Geology Research Centre (EGRU), College of Science and Engineering, James Cook University, Townsville, QLD, Australia

4 School of Earth Sciences and Engineering, Sun Yat-Sen University, Zhuhai 519000, Guangdong Province, China

5 Henan Academy of Geology, Zhengzhou 450001, Henan Province, China

Published online: 18 October 2025

Introduction

Sphalerite, one of the most common sulfide minerals, is the primary host mineral for Ga, Ge, Cd, and In in hydrothermal metallogenic systems (Tu et al. 2004; Cook et al. 2009; Gunn 2014; Frenzel et al. 2016; Bauer et al. 2018; Luo et al. 2022; Niu et al. 2024). According to previous research, In and Ga are incorporated into the sphalerite structure via a coupled substitution involving trivalent (Ga^{3+} and In^{3+}) and monovalent ions (such as Ag^{+} or Cu^{+}) (Johan 1988; Bauer et al. 2017; Cook et al. 2009; Ye et al. 2011; Frenzel et al. 2016; George et al. 2016; Cugerone et al. 2020; Xu et al. 2020, 2021; Graupner et al. 2024). Germanium occurs as Ge^{4+} and is incorporated into sphalerite via coupled substitution with monovalent elements such as Cu^{+} or through the creation of lattice vacancies (Belissont et al. 2016; Fougerouse et al. 2023; Luo et al. 2022; Liu et al. 2023; Torró et al. 2023). Cadmium always occurs as Cd^{2+} and is incorporated into sphalerite by substituting directly for Zn^{2+} or Fe^{2+} (Cook et al. 2009; Liu et al. 2010a; Belissont et al. 2014; Zhu et al. 2016). However, the concentration of these elements in sphalerite varies greatly between different hydrothermal deposit types (Tu et al. 2004; Cook et al. 2009; Gunn 2014; Frenzel et al. 2016; Bauer et al. 2018; Cugerone et al. 2020; Xu et al. 2021; Li et al. 2023a; Niu et al. 2024), suggesting that their content could be potential tracers of mineralization conditions and indicators of ore-forming fluid centers. Indium appears to be preferentially enriched in sphalerite from high-temperature deposits (Tu et al. 2004; Cook et al. 2009; Gunn 2014; Bauer et al. 2018), Ga and Ge are preferentially enriched in low-temperature deposits (Cook et al. 2009; Gunn, 2014; Belissont et al. 2014; Bauer et al. 2018), whereas Cd shows smaller concentration variations in sphalerite from different mineralization styles with a similar fluid source due to its similar geochemical behavior to Zn (Schwartz 2002; Cugerone et al. 2020). The behaviors of Ga, Ge, and Cd in magmatic-hydrothermal systems and their incorporation into sphalerite are far less understood compared to In. For instance, high concentrations (273–366 ppm) of Ga were reported in sphalerite from some epithermal deposits (Cook et al. 2009; Torró et al. 2023) but the factors affecting its transport in ore-forming fluids and its incorporation into the sphalerite structure are not fully explored. The behavior of these elements in other magmatic-hydrothermal systems, such as porphyry and skarn deposits, is still not well constrained.

The southern North China Craton (NCC) hosts the world's largest Mo resources and a variety of other deposit types, such as vein Au deposits, skarn Zn-Pb (Cu) deposits and vein Pb-Zn-Ag (Au) deposits (Fig. 1; Mao et al. 2008, 2011; Li et al. 2013, 2016, 2017; Zhao et al. 2018). There is broad consensus that the Yanshanian porphyry Mo deposits and their peripheral Pb-Zn-Ag deposits are the products of Early Cretaceous magmatic-hydrothermal activities and constitute typical magmatic-hydrothermal metallogenic systems, such as the Nannihu ore field (Ye 2006; Ye et al. 2006; Mao et al. 2011; Duan et al. 2010; Zhao et al. 2019; Li et al. 2023b). The Nannihu magmatic-hydrothermal system includes the Nannihu porphyry Mo-W deposit, Luotuoshan skarn Zn-Cu-W deposit, and Lengshuibeigou epithermal deposit, distributed from proximal to distal around the Nannihu intrusion (Li et al. 2023b). These systems offer an excellent opportunity to study the distribution of Ga, Ge, Cd, and In in magmatic-hydrothermal systems and to understand the factors that control the enrichment of these elements in sphalerite.

This study provides compositional data and fluid inclusion microthermometric results of sphalerite from three different deposits in the Nannihu ore field. These results are used to assess the potential control of temperature on trace element partitioning in magmatic-hydrothermal sphalerite. Our data are combined with analyses from other magmatic-hydrothermal deposits to evaluate the use of sphalerite trace-element geochemistry in locating the metallogenic center of magmatic-hydrothermal systems.

Geological background

The Nannihu ore field is located at the southern margin of the North China craton (Figs. 1 and 2a) and has resources of Mo (>2 Mt), W (640, 000 t of WO_3), Cu (63, 000 t), Zn + Pb (7 Mt) and Ag (8000 t) (Ye et al. 2006; Xiang et al. 2012; Cao et al.,2014 ; Yun et al., 2020; Li et al. 2023b). The Late Archean to Early Paleproterozoic Taihua Group constitutes the basement and is made up of metamorphic rocks which are composed mostly of amphibolite, felsic gneiss, migmatite, and metasedimentary rocks intercalated with mafic to ultramafic rocks (Hu et al. 1988). The basement is overlain by the Paleoproterozoic Xiong'er Group and Mesoproterozoic to Paleozoic metaclastic and littoral carbonate rocks. The Xiong'er Group comprises a thick sequence of volcanic rocks ranging in composition from basalt to rhyolite, with minor intercalations of clastic rocks (Zhao et al. 2002; Peng et al. 2008). The carbonate rocks overlie the Xiong'er Group in the south, and consist of the Guandaokou Group, Luanchuan Group, and Taowan Group from bottom to top (Fig.1).

Since the Early Mesozoic, the southern NNC and the Qinling orogen have undergone north-south directed compressional deformation during the orogenic period and the post-orogenic tectonic transition following the Qinling orogen related to the collision of the South China and North China cratons (Wu et al. 2005). During the late Mesozoic,

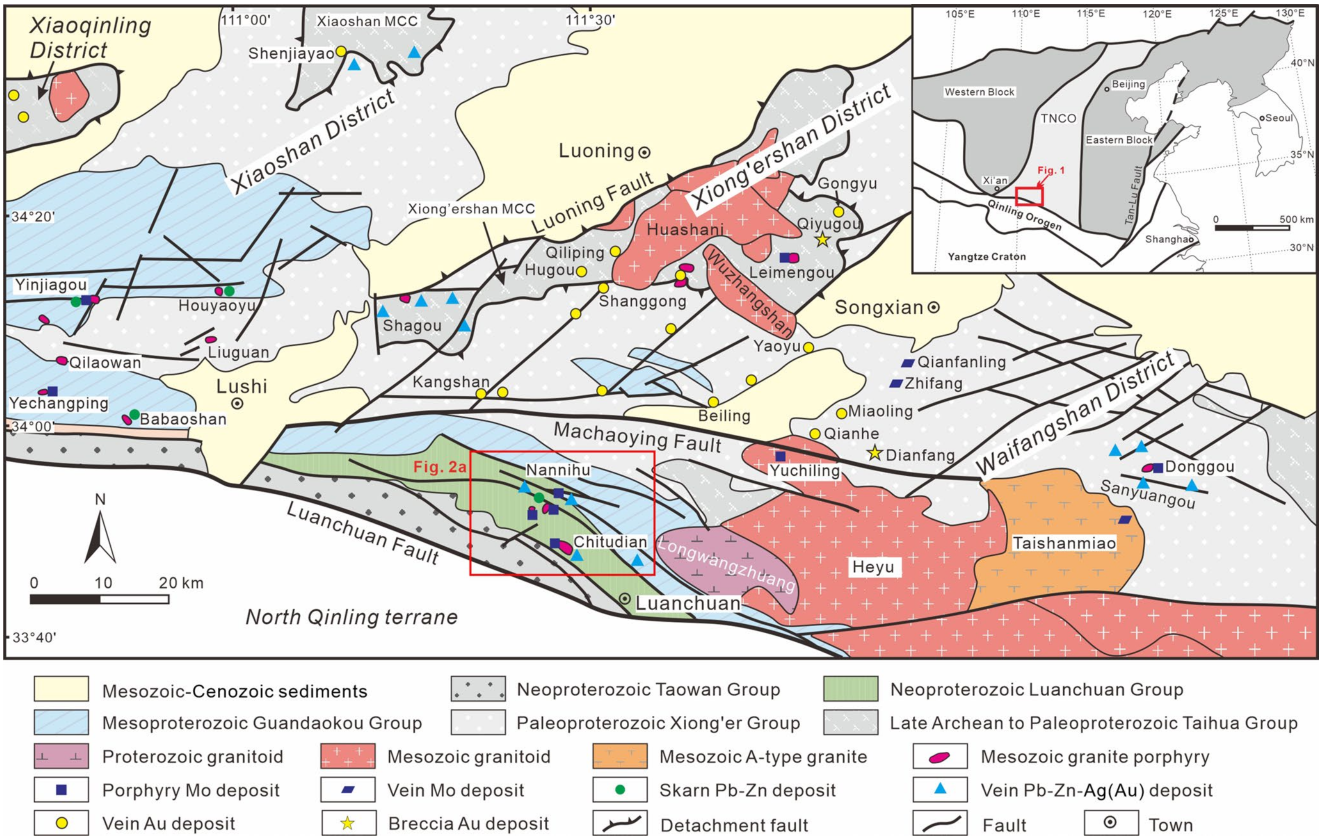


Fig. 1 Geological map showing major mineral districts along the southern margin of the NCC.(modified from Li et al. 2013). The insert map shows the tectonic location of the southern NCC in Eastern China. Abbreviations: TNCO: Trans-North China Orogen, MCC: metamorphic core complex

the southern NNC experienced lithospheric thinning related to back-arc extension resulting from the paleo-Pacific subduction beneath the Eurasian plate (Zhu et al. 2011). Multiple faults, magmatic activities, and mineralization events resulted from this multi-stage Mesozoic tectonic history (Zhao et al. 2019). The structural architecture is dominated by EW- and NNE-trending faults in the southern NNC. The EW-trending faults control the regional tectonic framework, and include the Sanbao Fault, the Machaoying Fault, and the Luanchuan Fault from north to south (Fig.1). By contrast, the NNE-trending faults are developed only locally and record a shorter duration of tectonic activity. However, these faults have a temporal and spatial relationship with regional mineralization and appear to control the occurrence of polymetallic deposits in the southern NCC (Fig. 1; Chen and Fu 1992; Mao et al. 2008, 2010; Yan et al. 2009; Li et al. 2012a, 2012b; Zhao et al. 2019).

The southern NCC experienced frequent and intensive granitoid intrusions along faults or between stratigraphic layers during the Mesozoic (Fig. 1). The Huashani pluton, located at the center of the Xiong'er district, has a SHRIMP zircon U-Pb age of 132.0 ± 1.6 Ma (Li 2005) and LA-ICP-MS zircon U-Pb ages of 127.6 ~ 129.3 Ma (Xiao et al. 2012). The Wuzhangshan pluton has a SHRIMP zircon U-Pb age of 156.8 ± 1.2 Ma (Li 2005) and an amphibole Ar-Ar age of 156.0 8 ± 1.1 Ma (Han et al. 2011). The Heyu pluton in the Waifangshan district is a composite batholith formed by multiple-phase intrusive magmatism during 148.2 ~ 127.2 Ma (Li 2005; Gao et al. 2010). The Taishanmiao pluton of the Waifangshan district is mainly composed of medium to coarse-grained syenite granite, with zircon U-Pb ages between 125 and 113 Ma (Ye et al. 2008; Gao et al. 2014; Wang et al. 2015). Moreover, several granitic porphyries associated with Mo-mineralization occur in this area with zircon U-Pb ages between 158 and 112 Ma (Fig. 1; Li 2005; Mao et al. 2010; Li et al. 2017, 2023b).

Geology of the Nannihu ore field

The Nannihu ore field is situated in the Luanchuan district, bounded by the Luanchuan fault in the south and by the Machaoying fault in the north (Figs. 1 and 2a). The ore field consists mainly of the Nannihu porphyry Mo-W deposit (0.66 Mt Mo at 0.06%, 0.14 Mt WO_3 at 0.081%, 0.3 Mt Re at 0.0027%), Shangfanggou porphyry Mo deposit (0.72 Mt Mo at 0.134%, 2.3 Mt Fe at 27.08%, 0.81 Mt Re at 0.0042%), Sandaozhuang skarn Mo-W deposit (0.67

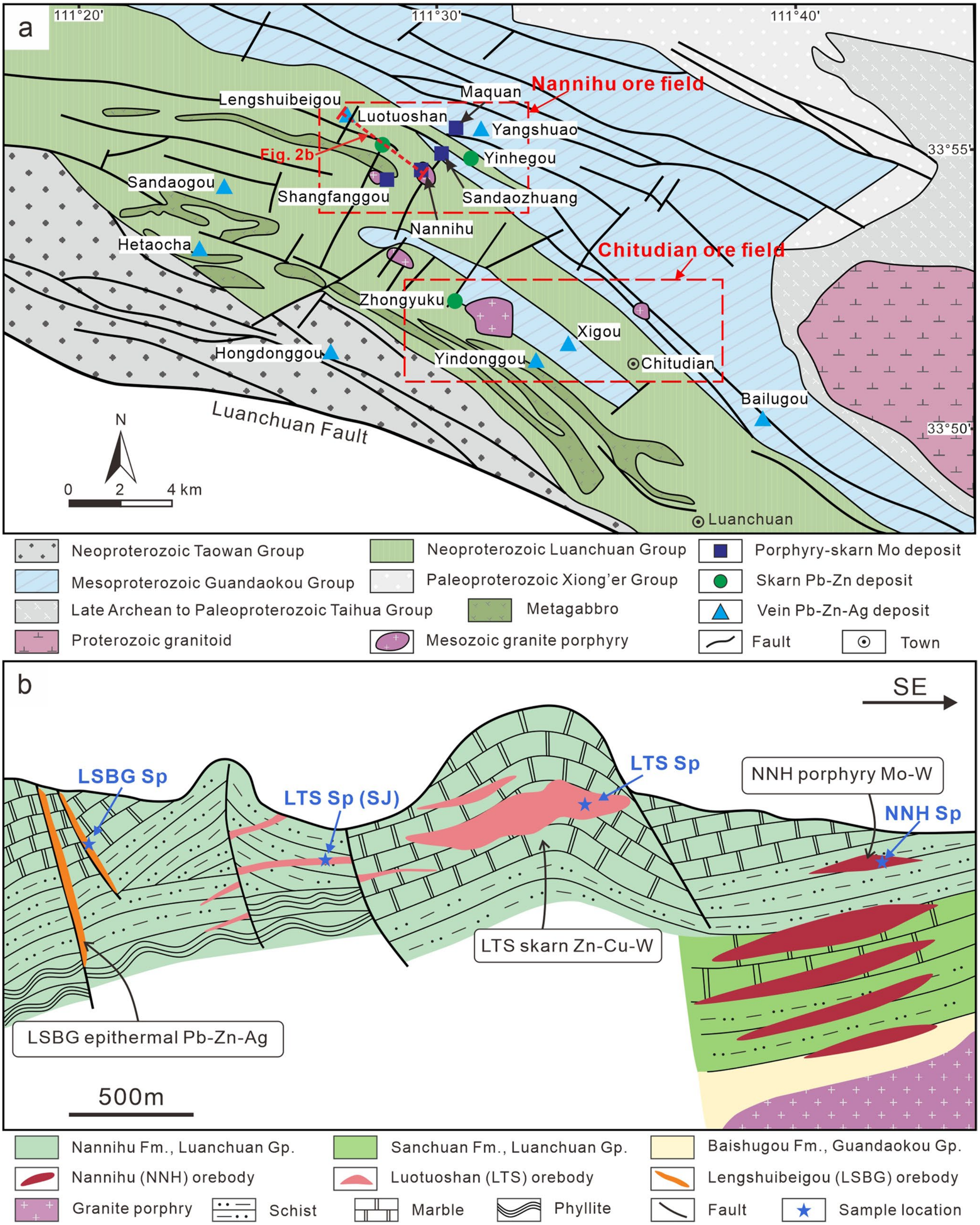


Fig. 2 (**a**) Geological map of the Luanchuan district (modified from Ye et al. 2006; Duan et al. 2010). (**b**) Geological section of the Nannihu porphyry Mo-W, Luotuoshan skarn Zn-Cu, and vein Lengshuibeigou Pb-Zn-Ag metallogenic system in the Nannihu ore field (modified from Zhang et al. 2023)

Mt Mo at 0.109%, 0.50 Mt WO_3 at 0.112%), Luotuoshan skarn Zn-Cu-W deposit (0.17 Mt Zn at 2.5%, 0.03 Mt Cu at 0.357%), the Lengshuibeigou epithermal Pb-Zn-Ag deposit (2.73 Mt Pb + Zn at 18.56%) (Fig. 2a; Guo 2006). Two main lithology groups are observed in the district: the Mesoproterozoic Guandaokou Group and the Neoproterozoic Luanchuan Group (Fig.2a). The Mesoproterozoic Guandaokou Group in the Nannihu ore field is dominated by the Baishugou Formation that is mainly composed of carbonate sediments with terrigenous clastic rocks. The Neoproterozoic Luanchuan Group uncomfortably overlies the Guandaokou Group and contains quartz sandstone, quartzite, marble, and dolomitic marble. The formations of the Luanchuan Group exposed in the ore field include the Sanchuan, Nannihu, and Meiyaogou Formations from the bottom to top (Fig. 2a).

The geological structures in the Nannihu ore field consist mainly of NW-trending folds and faults (Fig. 2a). The primary fold structures consist of the Qinghetang-Zhuangke anticline and the Baotouzhai-Nannihu syncline as the first order structures, and the Luotuoshan anticline and Shangfanggou syncline as the second order structures. In the ore field, NW- and NNE-trending faults are extensively developed. The Mesozoic granitoids were emplaced at the intersections of the NW-trending faults or the anticlinal core with the NNE-trending faults (Fig. 2a).

Magmatic rocks in the ore field include Neoproterozoic gabbro and Mesozoic granitoids (Fig. 2a). Several granitoid intrusions, emplaced in the Luanchuan Group, consist mainly of granite porphyry, porphyritic monzogranite, and granite dykes. The Nannihu intrusion is calc-alkaline and mineralogically consists of K-feldspar, biotite, quartz, and plagioclase. SHRIMP and LA-ICP-MS U-Pb dating on zircon from the Nannihu intrusion yielded emplacement ages of 157−145 Ma (Li 2005; Mao et al. 2010; Xiang et al. 2012; Li et al. 2023b).

The Nannihu (NNH) porphyry Mo-W deposit is located in the center of the Nannihu ore field, and its orebodies mainly occur in the contact zone between the Nannihu intrusion and the carbonate and schist wall rocks (Fig. 2b). The Luotuoshan (LTS) skarn Zn-Cu-W deposit, located to the north-west of the Nannihu deposit (Fig. 2a), consists of layered or lenticular-shaped orebodies in the skarn zone, which is hosted in the carbonate rocks of the Nannihu Formation. The Lengshuibeigou (LSBG) epithermal Pb-Zn-Ag deposit is in the northwestern part of the Nannihu district and is hosted by the Nannihu Formation along NW-trending faults (Fig. 2b). A detailed mineralization and alteration description of the three deposits is provided by Li et al. (2023b). Based on field and petrographic observations, three different paragenetic stages were recognized (Fig. 3). At the Nannihu porphyry deposit, the three paragenetic stages consist of: (1) Quartz-potassium feldspar stage, (2) Quartz-sulfide stage, and (3) Quartz-carbonate stage. At the Luotuoshan skarn deposit, the three paragenetic stages consist of: (1) Skarn stage, divided into prograde alteration and retrograde alteration, (2) Sulfide stage, and (3) Carbonate stage. At the Lengshuibeigou epithermal deposit, the three paragenetic stages consist of (1) Pyrite-quartz stage, (2) Quartz-polymetallic sulfide stage, and (3) Quartz-carbonate stage.

The Co/Ni ratios and trace element patterns of pyrite from all three deposits are consistent with a magmatic-hydrothermal origin and the metal zonation is typical of porphyry-related metallogenic systems (Li et al. 2023b). Co, Ni and other trace elements are enriched in the pyrite from the Nannihu deposit, while the pyrite from the Luotuoshan deposit exhibits relatively higher concentrations of Mn, Ni, and Bi. In contrast, the pyrite in the Lengshuibeigou deposit displays relatively higher concentrations of Pb, Zn, Cu, Au, Ag, As, and Sb (Li 2013; Li et al. 2023b). The $\delta^{34}S$ values of sulfide samples and H-O isotopes of quartz from the Nannihu deposit to the Lengshuibeigou deposit show a trend from magmatic fluids to meteoric waters (Yan 2004; Li 2013). Combined with the geochronological data, the Nannihu, Luotuoshan, and Lengshuibeigou deposits are recognized as the products of a large magmatic-hydrothermal system formed during 145−139 Ma (Li et al. 2023b).

Samples and analytical methods

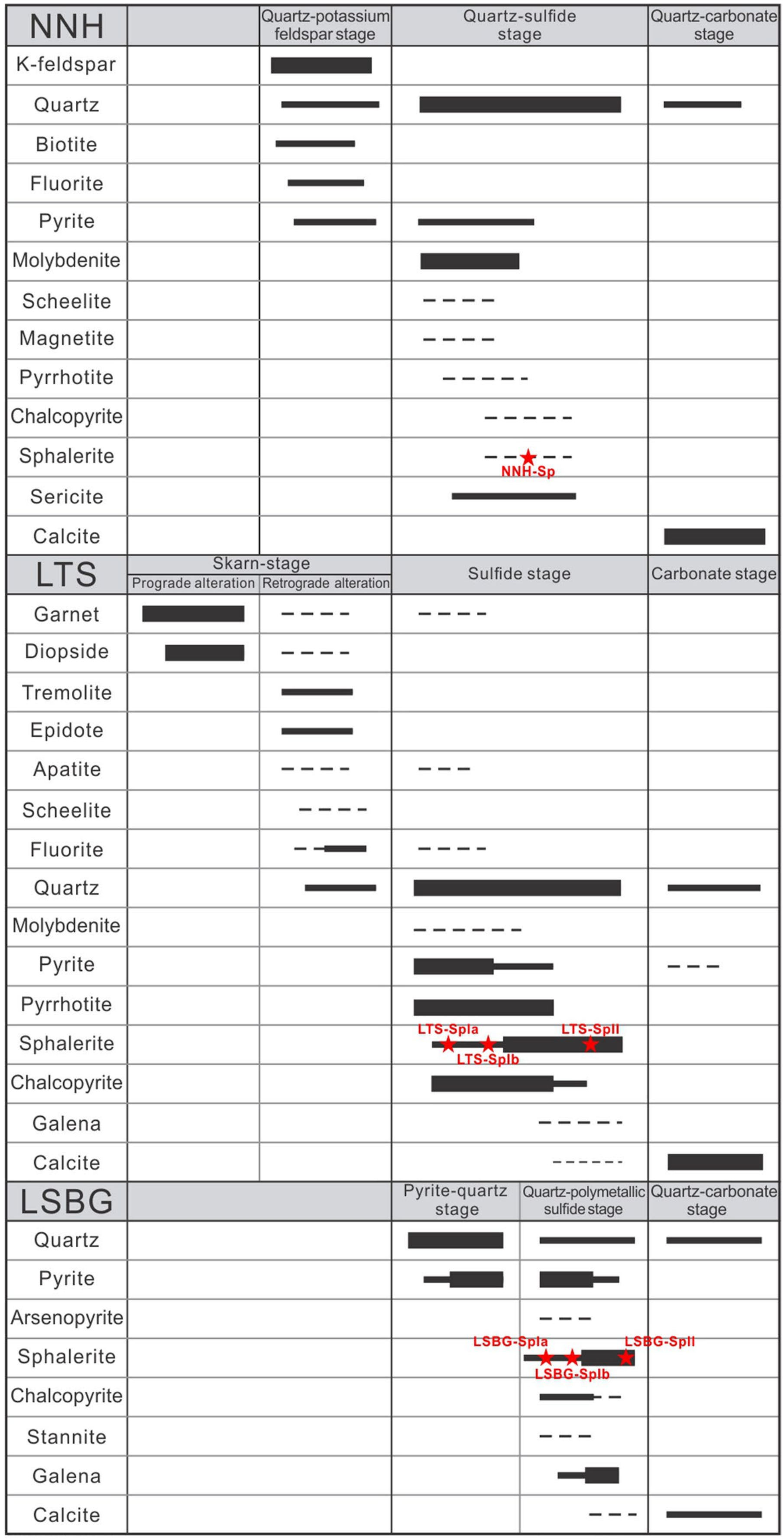
Sample descriptions

For this study, twelve representative samples of the Nannihu ore field were selected to investigate the composition of sphalerite from the Nannihu, Luotuoshan, and Lengshuibeigou deposits, and to characterize the fluids associated with sphalerite precipitation. These twelve samples were specifically chosen from a set of over 100 well-characterized samples from across all of these deposits, as they appropriately represent all the sphalerite types and are reasonably distributed in sampling locations in the Nannihu ore field (Fig. 2b).

Sample 21NNH-07 (Fig. 4a) was collected from the Nannihu deposit, which consists of K-feldspar, quartz, molybdenite, and pyrite, with minor amounts of rutile, magnetite, chalcopyrite, and sphalerite (Figs. 3a and 4). Sphalerite is closely associated with chalcopyrite in this sample and contains visible chalcopyrite inclusions (Fig. 4c-d). It formed later than K-feldspar, molybdenite, and other related minerals of quartz-potassium feldspar stage. No obvious internal zonation or growth texture are observed with NNH-Sp (Fig. 4c-d; ESM 2-Figure S1).

Seven samples (21LTS-06, 21LTS-02, 21LTS-13, 21LTS-19, 21LTS-21, SJ-03 and SJ-08) were collected from the

Fig. 3 Paragenetic sequence of minerals in the Nannihu (NNH) deposit, the Luotuoshan (LTS) deposit, and the Lengshuibeigou (LSBG) deposit (revised from Li et al. 2023b)



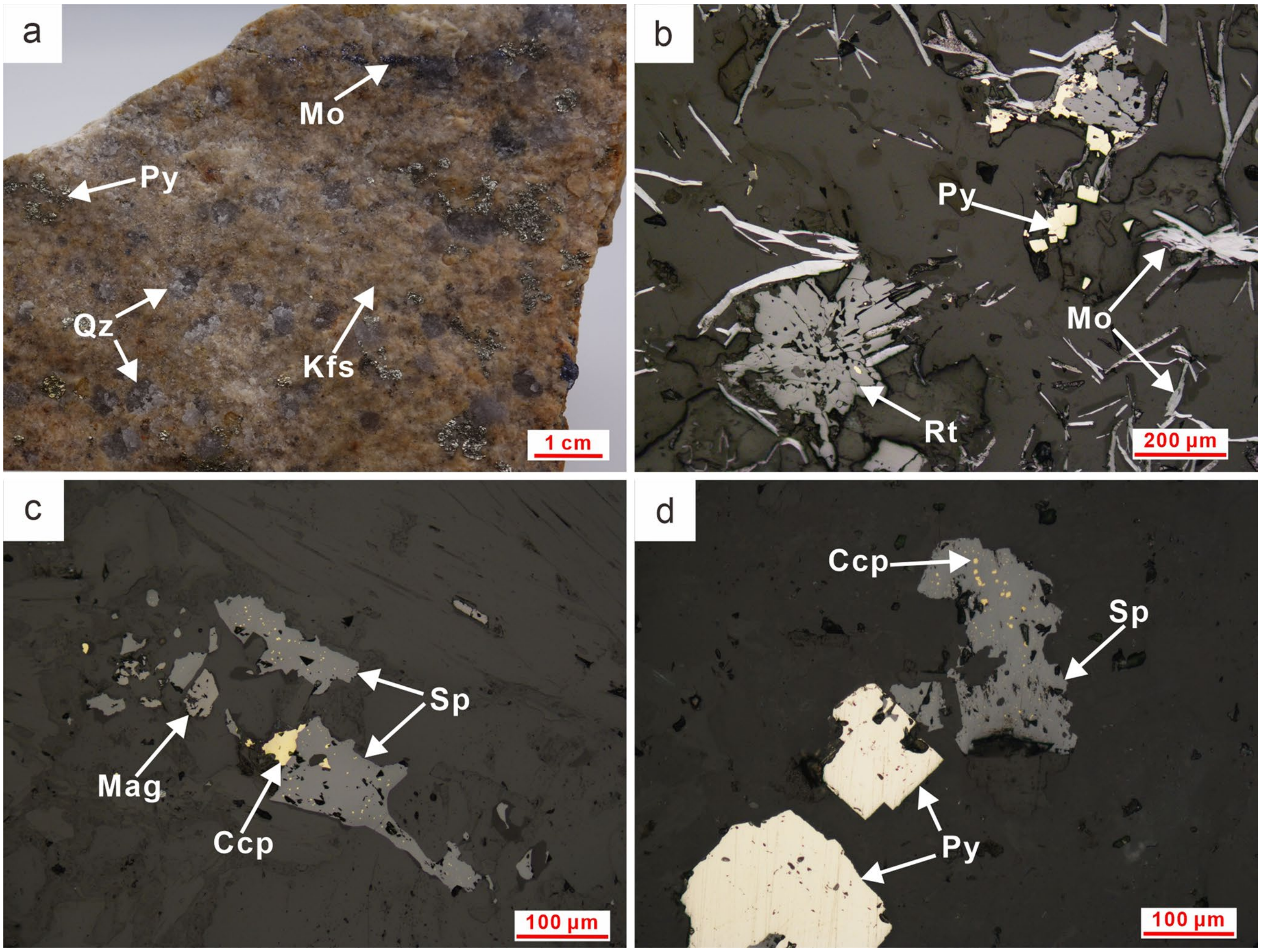


Fig. 4 Hand-specimen photo (a) and thin-section microphotographs (b-d) of sphalerite from the Nannihu Mo-W porphyry deposit. (**a**) Sample of Nannihu porphyry with K-feldspar, quartz, pyrite and molybdenite. (**b**) Subhedral pyrite replaced by rutile and scaly molybdenite. (**c**) Sphalerite replaced by chalcopyrite and containing chalcopyrite inclusions, with coexisting magnetite. (**d**) Pyrite and sphalerite. Irregular chalcopyrite inclusions occur in sphalerite. Abbreviations: Bt: biotite; Mol: molybdenite, Py: pyrite, Qz: quartz, Kfs: K-feldspar, Rt: rutile, Sp: sphalerite, Ccp: chalcopyrite, Mag: magnetite

Luotuoshan deposit. Sphalerite in the Luotuoshan deposit can be categorized into two types: LTS-SpI and LTS-SpII (Fig. 3b). LTS-SpI was collected from the sulfide-rich ore zone of the underground mine and can be subdivided into two subtypes: LTS-SpIa and LTS-SpIb. LTS-SpIa occurs in the samples 21LTS-06, SJ-03 and SJ-08, in association with pyrite, pyrrhotite, and chalcopyrite, and partially contains sparse chalcopyrite inclusions (Fig. 5a-c). LTS-SpIb appears in samples 21LTS-02 and 21LTS-13, and is characterized by sphalerite intergrowth with pyrrhotite (Fig. 5d). Minor chalcopyrite is rarely found within the fractures of LTS-SpIb (Fig. 5e), and occasionally occurs as linear trails of fine chalcopyrite grains distributed across large patches of sphalerite (Fig. 5f). LTS-SpIa and LTS-SpIb are distinguished by their different distributions on hand specimens (Fig. 5a, d) and associated mineral assemblages (Fig. 5b-c, e-f). LTS-SpII was obtained from the vicinity of the mine tunnel entrance from samples 21LTS-19 and 21LTS-21 (Fig. 5g), and occurs as infill between the skarn minerals (Fig. 5h-i). There is no other sulfide mineral associated with this type of sphalerite. Subtle growth zoning is locally visible in LTS-SpIa and LTS-SpIb, while LTS-SpII appears homogeneous with no obvious internal zoning (ESM 2-Figure S1).

Four samples (21LSBG-1-3, 21LSBG-1-2, 21LSBG-07, and 21LSBG-09) were collected from the Lengshuibeigou deposit. Based on their mineralogical features, sphalerite from these samples is categorized as LSBG-SpIa, LSBG-SpIb, and LSBG-SpII (Fig. 3c). LSBG-SpIa from samples 21LSBG-1-3 and 21LSBG-1-2 contains a large amount of chalcopyrite and tetrahedrite inclusions and is crosscut by LSBG-SpII veins (Fig. 6a). Chalcopyrite inclusions occur as oriented trails or as irregularly shaped grains, whereas the tetrahedrite inclusions are mainly coarse-grained

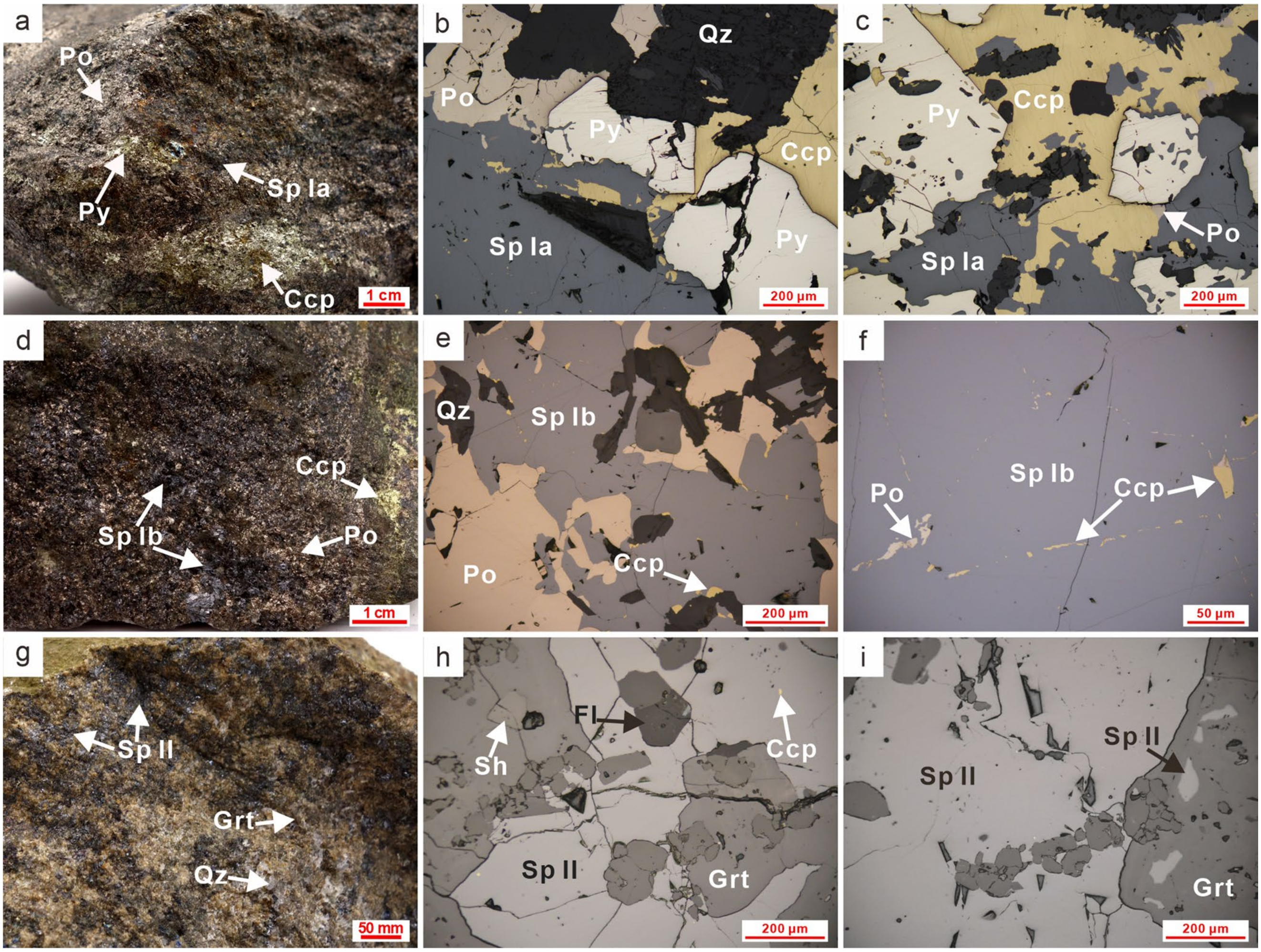


Fig. 5 Hand-specimen photo (a, d, g) and thin-section microphotographs (b-c, e-f, h-i) of sphalerite from the Luotuoshan (LTS) skarn deposit. (**a**) Massive sulfide ore sample with LTS-SpIa. Sphalerite, pyrite, pyrrhotite and chalcopyrite abundantly present in the ore sample. (**b**) Aggregates of sphalerite, pyrite, pyrrhotite, and chalcopyrite. LTS-SpIa replaces pyrite and pyrrhotite and is replaced by chalcopyrite, with chalcopyrite inclusions. (**c**) LTS-SpIa replacing pyrite and chalcopyrite. (**d**) Massive sulfide ore sample with LTS-SpIb. LTS-SpIb coexists with pyrite and pyrrhotite. (**e**) LTS-SpIb intergrown with pyrrhotite and interstitial quartz, with chalcopyrite occurring at the rim of sphalerite. (**f**) Large LTS-SpIb aggregate with pyrrhotite and chalcopyrite inclusions. (**g**) Ore sample containing sphalerite, garnet, and quartz. LTS-SpII fills in the space among garnet. (**h**) LTS-SpII coexisting with garnet, fluorite, and scheelite. (**i**) Sphalerite replacing garnet. Abbreviations: Py: pyrite, Sp: sphalerite, Po: pyrrhotite, Ccp: chalcopyrite, Qz: quartz, Grt: garnet, Fl: fluorite, Sh: scheelite

(Fig. 6c-e). LSBG-SpIb from sample 21LSBG-07 contains numerous inclusions of coarse chalcopyrite, and micron to sub-micron sized stannite (Fig. 6f-h). LSBG-SpII from samples 21LSBG-07 and 21LSBG-09 occurs as veins with pyrite on both margins (Fig. 6a-b). Compared to the homogeneous LSBG-SpIa and LSBG-SpIb, the coarse-grained LSBG-SpII exhibits growth textures (ESM 2-Figure S1).

Analytical methods

Backscattered electron (BSE) imaging via scanning electron microscopy (SEM) was used to investigate the internal textural features of each type of sphalerite (ESM 2-Figure S1). Major and minor elements of sphalerite were analyzed using a (JEOL) JXA-8230 electron probe microanalyzer (EPMA) at the Wuhan Sample Solution Analytical Technology Co., Ltd, Hubei, China. The operating conditions were set at an accelerating voltage of 20 kV and a beam current of 50 nA with a beam diameter of 5 µm. The crystals used were as follows: PETJ for Sn, In, Cd, and Ag; LIFH for Ga, Zn, Cu, Fe, and Mn; PETH for S.

Trace element analysis of sphalerite was conducted using an Agilent 7700e ICP-MS apparatus interfaced to a GeolasPro laser ablation system that consists of a 193 nm COMPexPro 102ArF excimer laser and a MicroLas optical system at the Wuhan Sample Solution Analytical Technology Co., Ltd. Hubei, China. Detailed operation conditions for the laser ablation system and the ICP-MS instrument

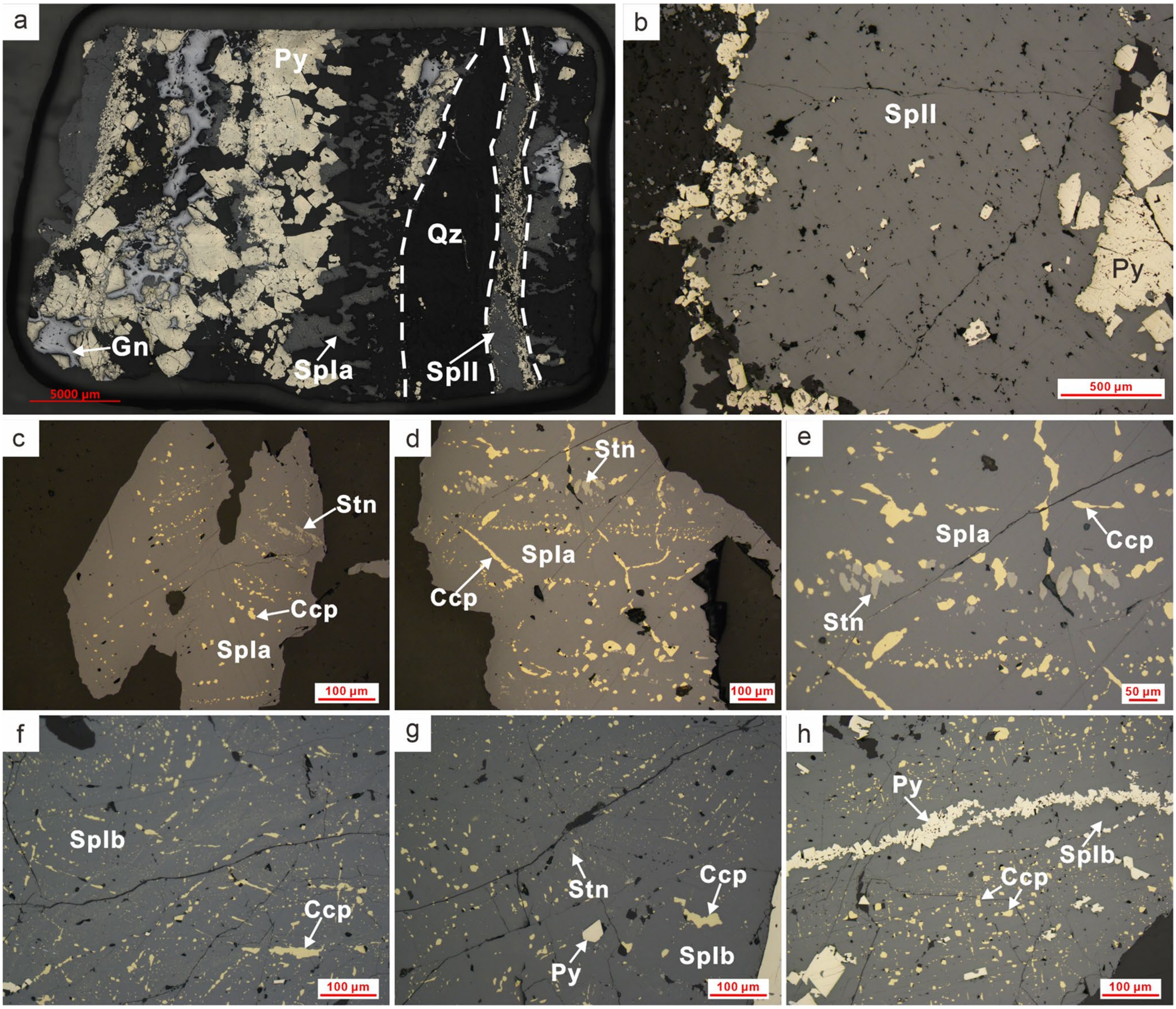


Fig. 6 Thin-section microphotographs of sphalerite from the Lengshuibeigou (LSBG) epithermal deposit. (**a**) LSBG-SpIa cut by LSBG-SpII vein with late quartz vein infill. Coarse-grained pyrite is replaced by sphalerite and galena. (**b**) LSBG-SpII replacing pyrite and surrounded by medium-grained pyrite. (**c**) LSBG-SpIa with irregular chalcopyrite and stannite inclusions. (**d**) LSBG-SpIa associated with chalcopyrite and stannite. Irregular stannite and chalcopyrite veinlets occur in LSBG-SpIa. (**e**) LSBG-SpIa intergrown with stannite and chalcopyrite. (**f**) LSBG-SpIb with large chalcopyrite inclusions and foggy submicron-sized inclusions. (**g**) LSBG-SpIb associated with stannite and chalcopyrite inclusions. LSBG-SpIb replaces the early euhedral-subhedral pyrite. (**h**) LSBG-SpIb replacing pyrite with chalcopyrite inclusions. Abbreviations: Sp: sphalerite, Stn: stannite, Ccp: chalcopyrite, Py: pyrite, Qz: quartz, Gn: galena

follow those described by Liu et al. (2008, 2010b). Laser ablation was performed using a 44 μm spot size, with the laser repetition rate set to 5 Hz and an energy density of 5.5 J/cm^2. Each analysis incorporated a background acquisition of approximately 20 to 30 s followed by a 50 s measurement on the sphalerite sample. Elements monitored during analysis include ^{55}Mn, ^{57}Fe, ^{63}Cu, ^{71}Ga, ^{72}Ge, ^{75}As, ^{107}Ag, ^{111}Cd, ^{115}In, ^{118}Sn, ^{121}Sb, ^{197}Au, ^{205}Tl. The Glass NIST SRM 610 was used as the external standard and MASS-1 sulfide reference material was used as the primary reference to calibrate trace element quantifications and identify possible instrumental drift. Zinc concentrations of sphalerite measured by EMPA were used as an internal calibration for LA-ICP-MS analysis.

Double-polished thick sections of samples 21LTS-06, 21LTS-13, 21LTS-19, 21LSBG-1, 21LSBG-07, and 21LSBG-09 were prepared for fluid inclusion petrography and microthermometry. Fluid inclusion petrography was performed under visible and infrared transmitted light using an Olympus infrared (IR) microscope coupled with a Qimaging Retiga-2000R digital CCD camera. Microthermometric measurements were conducted using a Linkam THMS

600 heating-freezing stage (−196 °C to 600 °C) at the State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences (Wuhan). The heating-freezing stage was calibrated at the melting points of CO_2, the ice-melting of H_2O, and the critical point of pure H_2O with synthetic fluid inclusion standards. The precision was estimated to be ±0.5 °C, ± 0.2 °C, and ±2 °C for the runs in the range of 120 °C to −70 °C, −70 °C to 100 °C, and 100 °C to 600 °C, respectively. To avoid the "warming effect" from IR light on microthermometric results of the fluid inclusions in sphalerite (Moritz 2006; Casanova et al. 2018; Peng et al. 2020), temperatures of ice melting (Tm) were measured using the improved cycling method (Peng et al. 2020). Moreover, homogenization temperatures (Th) were obtained under the lowest possible IR light intensity with the smallest diaphragm size to minimize the deviation (Moritz 2006; Peng et al. 2020). The salinities (wt % NaCl equiv) of fluid inclusions were calculated from the Tm, and the fluid density was determined using HokieFlincs (Steele-MacInnis et al. 2012).

Results

Major and trace elements of sphalerite

The EPMA results ($n=62$) indicate that sphalerite from different deposits from the Nannihu ore field has distinct Zn and Fe contents. The overall Zn content varies from 54.7 to 64.0 wt%, and the Fe content varies from 3.6 to 11.3 wt%. Sphalerite from the Nannihu deposit shows high average Zn contents of 62.3 wt% and low average Fe contents of 3.89 wt%. Sphalerite from the Luotuoshan deposit has the lowest average Zn content (55.3 wt% for LTS-SpIa, 55.1 wt% for LTS-SpIb, and 54.7 wt% for LTS-SpII), and the highest average Fe content (10.3 wt% for LTS-SpIa, 10.5 wt% LTS-SpIb, and 11.2 wt% for LTS-SpII). Sphalerite from the Lengshuibeigou deposit has similar average Zn (61.1 wt% for LSBG-SpIa, 62.9 wt% for LSBG-SpIb and 62.1 wt% for LSBG-SpII) and Fe (5.0 wt% for LSBG-SpIa, 3.6 wt% for LSBG-SpIb and 4.3 wt% for LSBG-SpII) composition to sphalerite from Nannihu deposit.

A total of 184 LA-ICP-MS spot analyses were conducted on sphalerite from the Nannihu ore field. The results are shown in ESM 1-Table S1 and the concentrations of selected trace elements are illustrated in Fig. 7. The sphalerite from the three studied deposits from the Nannihu ore field generally shows high contents of Cd and In, low Ge contents and variable Ga contents (Fig. 7).

Sphalerite from the Nannihu porphyry deposit (NNH-Sp) yields Ga contents of 0.64–7.26 ppm, Cd contents of 1426–1687 ppm, In contents of 291–460 ppm, and the Ge content is less than 2 ppm or below the detection limit (ESM 2-Figure S2). The Mn content (54.4–2515 ppm) is highly variable while the amounts of Ag (<10 ppm) and Sn (<5 ppm) are relatively low. Due to the presence of numerous chalcopyrite inclusions, the Cu content varies greatly compared to the other elements, e.g., Zn, Mn, Fe, Cd, and In (ESM 2-Figure S3a-b). For Cu, the LA-ICPMS signal can be divided into spikes corresponding to chalcopyrite inclusions, and more flat signals corresponding to Cu incorporated into the sphalerite structure. Processing these segments separately, it reveals that the pronounced peaks lead to elevated Cu contents, whereas Ga, Ge, In, and Cd contents remain relatively stable (ESM 2-Figure S3a-b; ESM 1-Table S2). This indicates that Ga, In, and Cd may occur within the sphalerite lattice rather than within chalcopyrite inclusions.

Sphalerite from the Luotuoshan skarn deposit shows a decrease in Mn content from LTS-SpIa (6086–8187 ppm) to LTS-SpIb (4619–5384 ppm) and to LTS-SpII

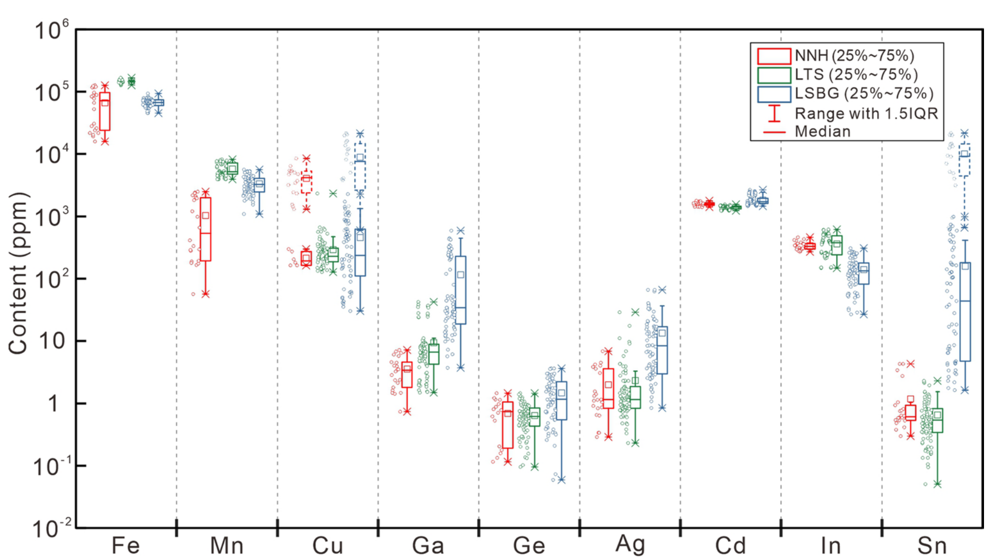


Fig. 7 Boxplot of LA-ICP-MS data for selected elements in sphalerite from the Nannihu (NNH), Luotuoshan (LTS), Lengshuibeigou (LSBG) deposits. There are 22 measurements for the Nannihu deposit, 87 measurements for the Luotuoshan deposit, and 75 measurements for the Lengshuibeigou deposit. The dashed outlines indicate data affected by chalcopyrite or stannite inclusions

(3929–4934 ppm) (Fig. 8a). The copper content shows a moderate increase from LTS-SpIa (128–640 ppm) to LTS-SpIb (244–664 ppm) whereas the Cu content of LTS-SpII (187–564 ppm) shows moderate contents and the least variability (Fig. 8b). The In content shows similar variations to Cu, increasing from LTS-SpIa (147–270 ppm) to LTS-SpIb (452–616 ppm) and with moderate amounts in LTS-SpII (354–419 ppm) (Fig. 8e). Gallium contents decrease from LTS-SpIa to LTS-SpIb and to LTS-SpII respectively (Fig. 8c). The Cd content is high in all three types of sphalerite and decreases from LTS-SpIa (1369–1558 ppm) to LTS-SpIb (1339–1458 ppm) and to LTS-SpII (1234–1344 ppm), respectively (Fig. 8d). The Ag content is less than 30 ppm whereas the Ge and Sn concentrations are less than 2 ppm in all types of sphalerite from this deposit.

Manganese concentrations in sphalerite from the Lengshuibeigou epithermal deposit range from 2474 to 5340 ppm in LSBG-SpIa, 1899–4525 ppm in LSBG-SpIb, and 1096–5628 ppm in LSBG-SpII (Fig. 8g). Because of numerous stannite micro-inclusions, LSBG-SpIb has highly variable Cu and Sn contents (145-21562 ppm Cu and 2.94–21748 ppm Sn), with a ratio of ~2:1 and positively correlated (Fig. 8m). By comparison, LSBG-SpIa and LSBG-SpII contain relatively low Cu (240-12760 ppm, 30.3–496 ppm) and Sn concentrations (18.6–983 ppm, 1.63–38.2 ppm). LSBG-SpIa has very high Ga content (140–589 ppm), whereas the Ga content in LSBG-SpIb is low (3.71–52.4 ppm) and in LSBG-SpII (9.81–119 ppm) is moderate (Fig. 8h). The Ag content (Fig. 8i) varies between the three types of sphalerite (0.84–8.55 ppm for LSBG-SpIa; 3.11–66.4 ppm for LSBG-SpIb; and 2.06-21.0 ppm for LSBG-SpII). The Cd content (Fig. 8j) is in general high and variable (1623–1800 ppm for LSBG-SpIa; 1749–2673 ppm for LSBG-SpIb; and 1453–1756 ppm for LSBG-SpII). The In content (Fig. 8k) is moderate to high (101–272 ppm for LSBG-SpIa; 26.7–308 ppm for LSBG-SpIb; and 29.0-261 ppm for LSBG-SpII).

Petrography of fluid inclusions

No fluid inclusions were found in sphalerite samples collected from the Nanihu deposit. The fluid inclusions in sphalerite and quartz from the Luotuoshan and Lengshuibeigou deposits are dominated by two-phase liquid-rich inclusions and subordinate vapor-rich inclusions (Fig. 9). They occur as isolated fluid inclusion or as petrographically well-defined fluid inclusion assemblages (FIAs). Most fluid inclusions exhibit rectangular, rhombic, and negative-crystal shapes (Fig. 9).

The fluid inclusions in LTS-SpIa are rectangular and generally smaller than 5 μm, with a liquid/vapor ratio of about 1:1 (Fig. 9a). The quartz coexisting with LTS-SpIa is named LTS-QzIa. The LTS-QzIa quartz fluid inclusions are dominated by irregular or negative-crystal inclusions. They are usually liquid-rich and 2 to 5 μm in size (Fig. 9b-c).

No fluid inclusions were identified in LTS-SpIb. The LTS-SpII is dominated by rectangular or negative-crystal inclusions which occur as homogeneous FIAs with similar liquid/vapor ratios of about 6:4 (Fig. 9d-g).

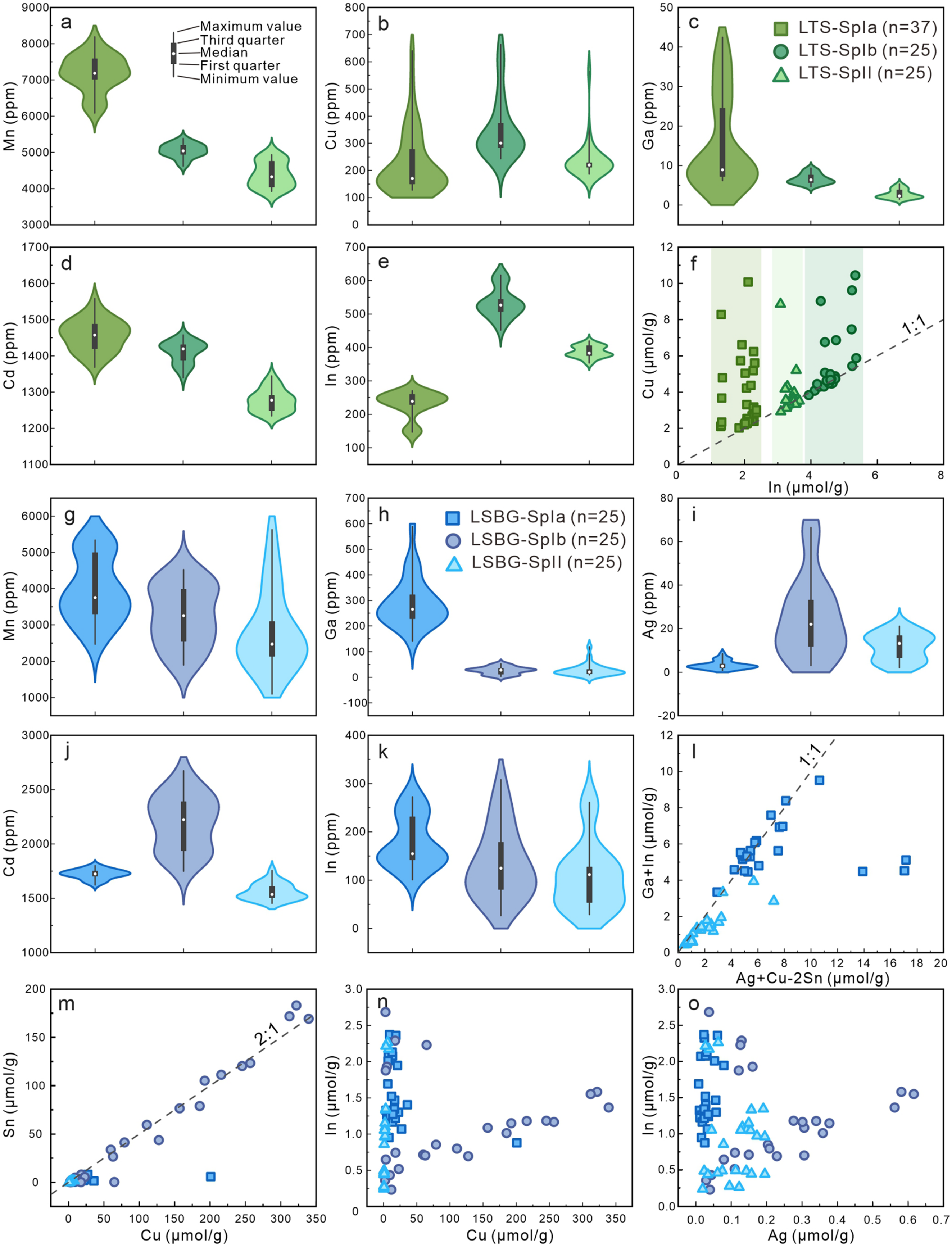
The LSBG-SpIa fluid inclusions are uncommon and show irregular or rectangular shapes with a size of 10 to 20 μm (Fig. 9h-i). Several liquid-rich inclusions are found in LSBG-SpIb (Fig. 9j), and in associated quartz (LSBG-QzIb) grains (Fig. 9k). Some rectangular or irregular inclusions were also found in LSBG-SpII (Fig. 9l).

Microthermometric measurements

A total of 74 fluid inclusions were studied from 16 FIAs and 10 isolated fluid inclusions. The microthermometric results and calculated parameters are summarized in ESM 1-Table S3 and graphically illustrated in Fig. 10. Based on observation under visible and infrared transmitted light, no measurable fluid inclusions in NNH-Sp and LTS-SpIb were identified.

No precise T_m of fluid inclusions from LTS-SpIa was obtained, but T_h varies from 402 to 396 °C (Fig. 10a). Fluid inclusion assemblages in LTS-QzIa have T_m of −5.5 to −2.4 °C and calculated salinities of 4.0 to 8.6 wt% NaCl equiv, with T_h ranging from 360 to 329 °C (Fig. 10a, b). The calculated fluid densities show a range from 0.711 to 0.730 g/cm^3. The FIAs from LTS-SpII record T_m values of −10.1 to −5.3 °C, which correspond to calculated salinities of 8.3 to 14.0 wt% NaCl equiv. (Fig. 10b). Their T_h values vary from 328 to 283 °C and fluid densities are calculated at 0.802 to 0.853 g/cm^3 (Fig. 10a).

The isolated fluid inclusions in LSBG-SpIa show Th of 329 to 297 °C (Fig. 10a). The FIAs in LSBG-QzIa have T_m of −4.7 to −1.7 °C and calculated salinities of 2.5 to 7.4 wt% NaCl equiv (Fig. 10a). Their total Th varies from 361 to 321 °C and calculated fluid densities range from 0.665 to 0.707 g/cm^3 (Fig. 10a). In LSBG-SpIb, the fluid inclusions show T_h of 290 to 286 °C (Fig. 10a). The FIAs from LSBG-SpII have T_m of −4.2 to −1.8 °C and T_h of 243 to 228 °C, which are similar to the T_m (−2.8 to −4.3 °C) and T_h (246 to 236 °C) of the isolated fluid inclusions (Fig. 10a). The calculated salinities and fluid densities of these FIAs are 3.1 to 6.7 wt% NaCl equiv. and 0.854 to 0.868 g/cm^3 respectively (Fig. 10b). The isolated fluid inclusions have calculated salinities of 4.6 to 6.9 wt% NaCl equiv. and fluid densities of 0.843 to 0.871 g/cm^3 (Fig. 10b).



a
b
c
d
e
f
g
h
i
j
k
l
m
n
o
Maximum value
Third quarter
Median
First quarter
Minimum value
LTS-SpIa (n=37)
LTS-SpIb (n=25)
LTS-SpII (n=25)
LSBG-SpIa (n=25)
LSBG-SpIb (n=25)
LSBG-SpII (n=25)
Mn (ppm)
Cu (ppm)
Ga (ppm)
Cd (ppm)
In (ppm)
Cu (μmol/g)
In (μmol/g)
Ag (ppm)
Ga+In (μmol/g)
Ag+Cu-2Sn (μmol/g)
Sn (μmol/g)
Cu (μmol/g)
In (μmol/g)
Ag (μmol/g)
1:1
2:1

Fig. 8 (**a-e**) Violin plots for selected trace and minor elements in LTS-SpIa, LTS-SpIb and LTS-SpII. (**f**) Correlation plot for Cu vs. In in LTS Sp. (**g-k**) Violin plots for selected trace and minor elements in LSBG-SpIa, LSBG-SpIb and LSBG-SpII. (**l-o**) Correlation plots for Ga + In vs. Ag + Cu-2Sn, Sn vs. Cu, In vs. Cu, In vs. Ag in LSBG Sp

Discussion

Occurrence and substitution of trace elements in sphalerite

Sphalerite is considered an important host mineral for a wide range of minor and trace elements, especially the dispersed elements such as Ga, Ge, In, and Cd (Cook et al. 2009; Murakami and Ishihara 2013; Belissont et al. 2014; Xu et al. 2021; Torró et al. 2023). LA-ICP-MS analysis can provide significant information about the occurrence of a given element in minerals. If elements are present as a micro-inclusion of a different mineral, this would be visible on the ablation profile (e.g., Ye et al. 2011). Lattice-bound trace elements and submicroscopic inclusions or nanoparticles will exhibit flat curves in the signal spectra, making them difficult to distinguish (e.g., George et al. 2015). As mentioned above, many mineral inclusions occur in sphalerite from the Nannihu and Lengshuibeigou deposits. In NNH-Sp and LSBG-SpIa, chalcopyrite inclusions are identified by abnormal Cu signal curves (ESM 2-Figure S3a-d). In LSBG-SpIb, the signal spectra show a smooth profile, even though the area selected for analysis is characterized by the presence of numerous submicron inclusions (ESM 2-Figure S3e).

The high Cu concentrations in the Nannihu and Lengshuibeigou deposits (Fig. 7) are coincident with the observation of inclusions in Figs. 4 and 6. There are obvious chalcopyrite inclusions in sphalerite from the Nannihu deposit so that a direct plot can neither avoid the effect of inclusions nor show the relationship between In and Cu. Sphalerite unaffected by chalcopyrite inclusions from the Nannihu deposit (ESM 1-Table S2) shows a narrow range of trace elements concentrations and a positive correlation between In and Cu, similar to that observed in the sphalerite from the Luotuoshan deposit (Fig. 8f). In the Lengshuibeigou deposit, there also could be a close relationship between In and Cu in sphalerite without inclusions (Fig. 8l). This indicates that In is incorporated into the studied sphalerite mainly via the couple substitution of Cu^{+}+ In^{3+}↔ $2Zn^{2+}$.

In the Lengshuibeigou deposit, Sn in sphalerite occurs most likely as Sn^{2+} due to the presence of stannite. According to the possible solid solutions in the Zn-(Cu + Ag)-(Sn + In) system (Ohta 1995), Sn can exist as Cu_2ZnSnS_4 sphalerite-stannite solid solution. Gallium, as Ga^{3+}, can enter sphalerite lattices in the same way as In^{3+} (Cook et al. 2009; Ye et al. 2011; Frenzel et al. 2016; Cugerone et al. 2020). There is a good correlation between In + Ga and Ag + Cu-2Sn and most analyses plot close to the 1:1 line (Fig.8l), indicating the operation of the coupled substitution Ag^{+} or Cu^{+} + In^{3+} or Ga^{3+} ↔ $2Zn^{2+}$ in LSBG-SpIa and LSBG-SpII.

Characterized by abundant micro-inclusions or submicron-inclusions of chalcopyrite and stannite, LSBG-SpIb contains diverse apparent contents of Cu, Sn, and Ag (Fig. 8i, m), with strong positive correlations between Sn and Cu, In and Cu, In and Ag (Fig. 8m, n, o). The LA-ICP-MS depth profile (ESM 2-Figure S3f) reveals that stannite inclusions can cause irregularities on Sn, Cu, In, and Ag, implying that In and Ag can be incorporated into stannite as $(Cu, Ag)_2Fe(Sn, In)S_4$. The In content in LSBG-SpIb is similar to that in LSBG-SpII (Fig. 8k), and there is a small peak with comparable amplitude in the signal for In (ESM 2-Figure S3f), indicating that stannite is not the main carrier of indium.

Temperature indication of trace elements in sphalerite

The composition of sphalerite can provide information about fluid temperature (Keith et al. 2014). Several studies have used the trace element content of sphalerite to estimate the formation temperature (Frenzel et al. 2016; Zhang et al. 2022; Zhao et al. 2024). The temperature range calculated using trace elements based on GGIMFis geothermometer for LTS-SpIa, LTS-SpIb, LSBG-SpIa and LSBG-SpIb are 427 ~ 331 °C, 428 ~ 337 °C, 333 ~ 257 °C and 395 ~ 280 °C, respectively, and these encompass the homogenization temperatures of the fluid inclusions in these sphalerite samples (Fig.10e). The numerous chalcopyrite and stannite inclusions in LSBG-SpIb may interfere with the temperature calculation, resulting in a high upper limit of the temperature range (Figs. 6f-h, 8m and 10e). Nonetheless, the temperatures calculated by GGIMFis for LTS-SpII and LSBG-SpII (445 ~ 351 °C, 385 ~ 285 °C) differ significantly from the homogenization temperatures (328 ~ 283 °C, 243 ~ 228 °C) of fluid inclusions (Fig. 10e). The former formed without the simultaneous growth of other sulfides, while the latter, except for replacing early pyrite, does not coexist with other sulfides such as chalcopyrite or stannite (Figs. 5h-i and 6b). Hence, it is worth noting that when using the GGIMFis geothermometer, the presence of associated minerals with sphalerite should be taken into account.

Based on the FA geothermometer (Zhang et al. 2022), the calculated temperatures for LTS-SpII and LSBG-SpII (339 ~ 304 °C, 307 ~ 255 °C) are close to the homogenization temperatures of the relevant fluid inclusions. The temperature ranges calculated by the SPRFT thermometer (Zhao et al. 2024) for LTS-SpII and LSBG-SpII (307 ~ 255 °C, 297 ~ 245 °C) also partially overlap the fluid inclusion

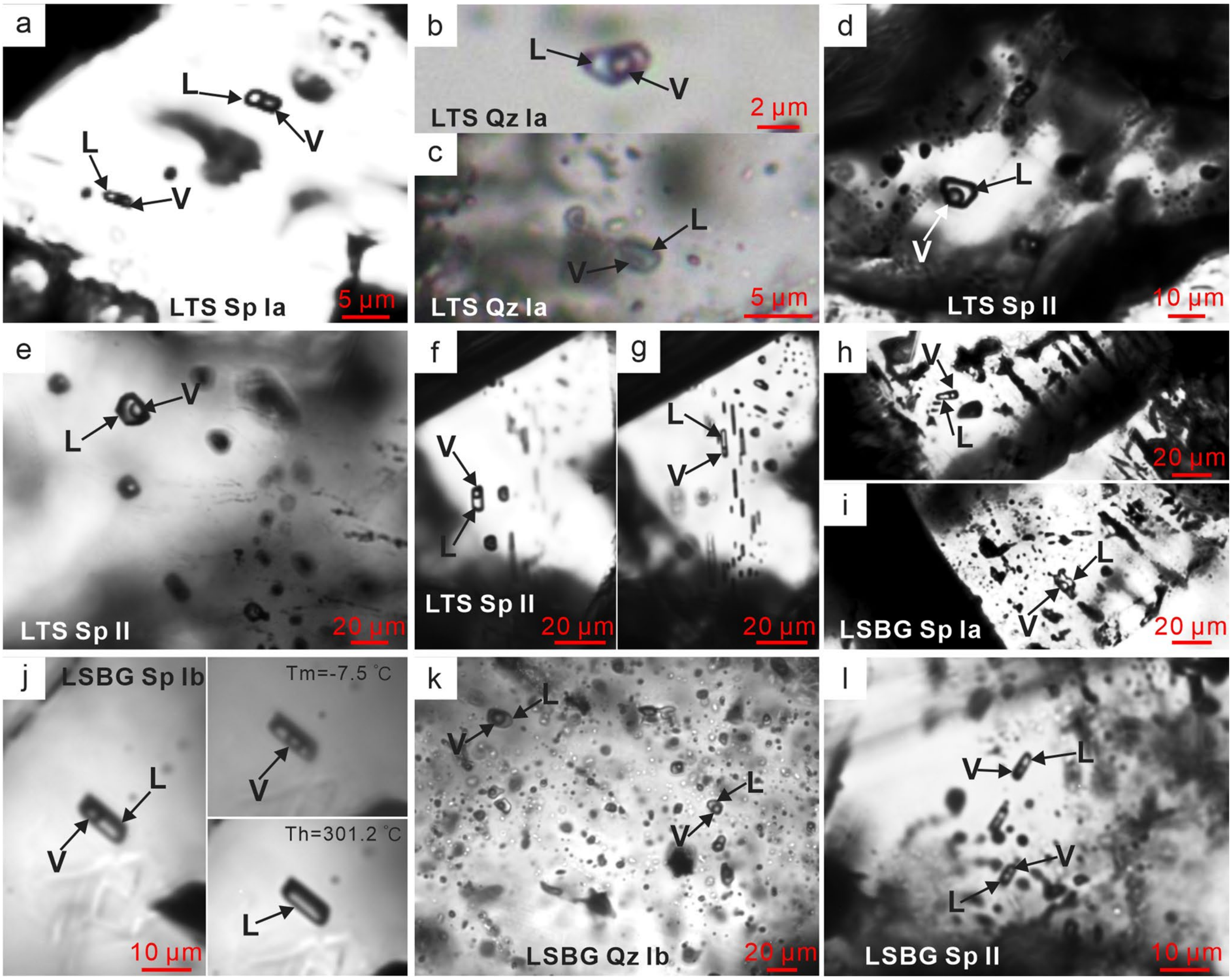


Fig. 9 Fluid inclusions in sphalerite and quartz. (**a**) Infrared photomicrograph of liquid-vapor fluid inclusions in LTS-SpIa. (**b-c**) Photomicrograph of liquid-vapor fluid inclusions in LTS-QzIa. (**d-g**) Infrared photomicrograph of liquid-vapor fluid inclusions in LTS-SpII. (**h-i**) Infrared photomicrograph of liquid-vapor fluid inclusions in LSBG-SpIa. (**j**) Infrared photomicrograph the ice temperature and homogenization temperature of liquid-vapor fluid inclusions in LSBG-SpIb. (**k**) Photomicrograph of liquid-vapor fluid inclusions in LSBG-QzIb. (**l**) Infrared photomicrograph of liquid-vapor fluid inclusions in LSBG-SpII

homogenization temperature range (Fig.10f). However, the temperature ranges obtained from these two geothermometers for LTS-SpIa, LTS-SpIb and LSBG-SpIa clearly deviate from the homogenization temperatures (Fig. 10f). Given that the sphalerite temperatures for the simulation of the FA and SPRFT geothermometer primarily range from 100～300 °C (Zhang et al. 2022; Zhao et al. 2024), there may be errors when predicting higher homogenization temperatures of sphalerite. Therefore, it is essential to select an appropriate geothermometer to estimate the formation temperature of sphalerite, and if necessary, the results from different geothermometers should be integrated. In general, the GGIMFis geothermometer provides more accurate temperature estimates when sphalerite is accompanied by several paragenetically associated minerals such as chalcopyrite and pyrrhotite, and its applicable temperature range may be broader.

In order to figure out the distribution characteristics of trace elements (Ga, Ge, In, and Cd) and their potential use in the exploration of magmatic-hydrothermal systems, additional sphalerite data from 44 different Zn-Pb deposits, including 11 skarn deposits, 10 MVT deposits, 10 VMS deposits, 10 epithermal deposits, and 3 SEDEX deposits, have been compiled from the literature (Fig. 11; ESM 1-Table S4). Gallium concentration displays a wide range and is inversely correlated to temperature (Fig. 11a). Compared to other deposits, high Ga contents appear to be more common in sphalerite from MVT deposits. It is worth noting that some analyses from epithermal deposits show very high Ga concentrations although the overall Ga concentration in

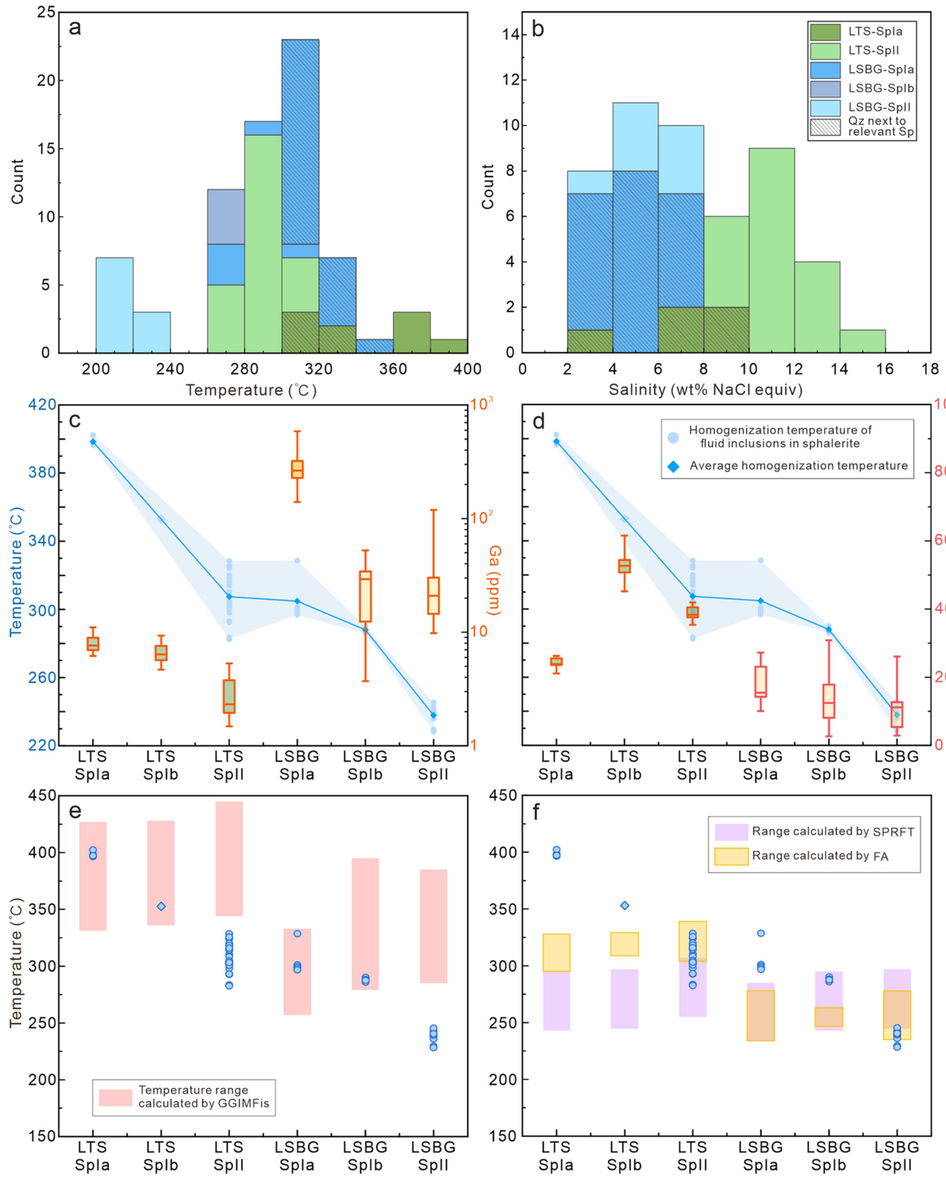


Fig. 10 (**a-b**) Histograms of homogenization temperature and calculated salinities of fluid inclusions from sphalerite and quartz. (**c-d**) Dual-axis plots illustrating the variation in temperature (left axis) and the concentrations of Ga or In (right axis) across different sphalerite, with the blue line connecting the average homogenization temperatures for each sphalerite type. For each sphalerite type, 25 measurements were utilized to construct the box plot. (**e-f**) Comparison of homogenization temperatures of fluid inclusions in sphalerite with results from GGIMFis geothermometer (T= -(54.4±7.3)* $\ln\left(\frac{C_{Ga}^{0.22}C_{Ge}^{0.22}}{C_{Fe}^{0.37}C_{Mn}^{0.20}C_{In}^{0.11}}\right)$+ (208±10)) (Frenzel et al. 2016), FA geothermometer (Zhang et al. 2022) and SPRFT geothermometer (Zhao et al.,2024). The homogenization temperature of LTS-SpIb is the mean of average homogenization temperatures of LTS-SpIa and LTS-SpII

these deposits is low. The high Ga content (up to 451 ppm) in LSBG-SpIa taken together with the moderate to low Ga contents in other sphalerite samples (Fig. 8c) from this area are consistent with this pattern.

The Ge distribution indicates that Ge-rich sphalerite tends to occur mainly in MVT deposits (Fig. 11b). In skarn and VMS deposits, Ge contents are low to moderate, and most sphalerite samples yield low Ge concentrations (Fig. 11b). Deposits associated with magmatic-hydrothermal systems are more likely to have limited potential for Ge. Most deposits have high Cd contents, except MVT deposits which yield highly variable Cd values (Fig. 11c). High In concentrations are common in skarn and epithermal deposits (Fig. 11d) whereas some samples from VMS deposits show high In concentrations, indicating the possibility of In enrichment in those deposits.

Sphalerite from magmatic-hydrothermal systems shows higher Ga and In contents as revealed by our compilation of 1639 LA-ICP-MS analyses of sphalerite from these deposits. The spatial distribution of trace elements in these systems can be a function of the distance between the ore deposition and the causative intrusion. Close to the causative intrusion, the presence of high-temperature fluids with a strong magmatic affinity is favorable for In enrichment.

Spatial distribution of dispersed elements in the magmatic-hydrothermal system

At the scale of the Nannihu ore field, In distribution appears to be controlled by temperature and proximity to the magmatic center. Sphalerite from the higher-temperature and proximal Luotuoshan deposit has higher In contents

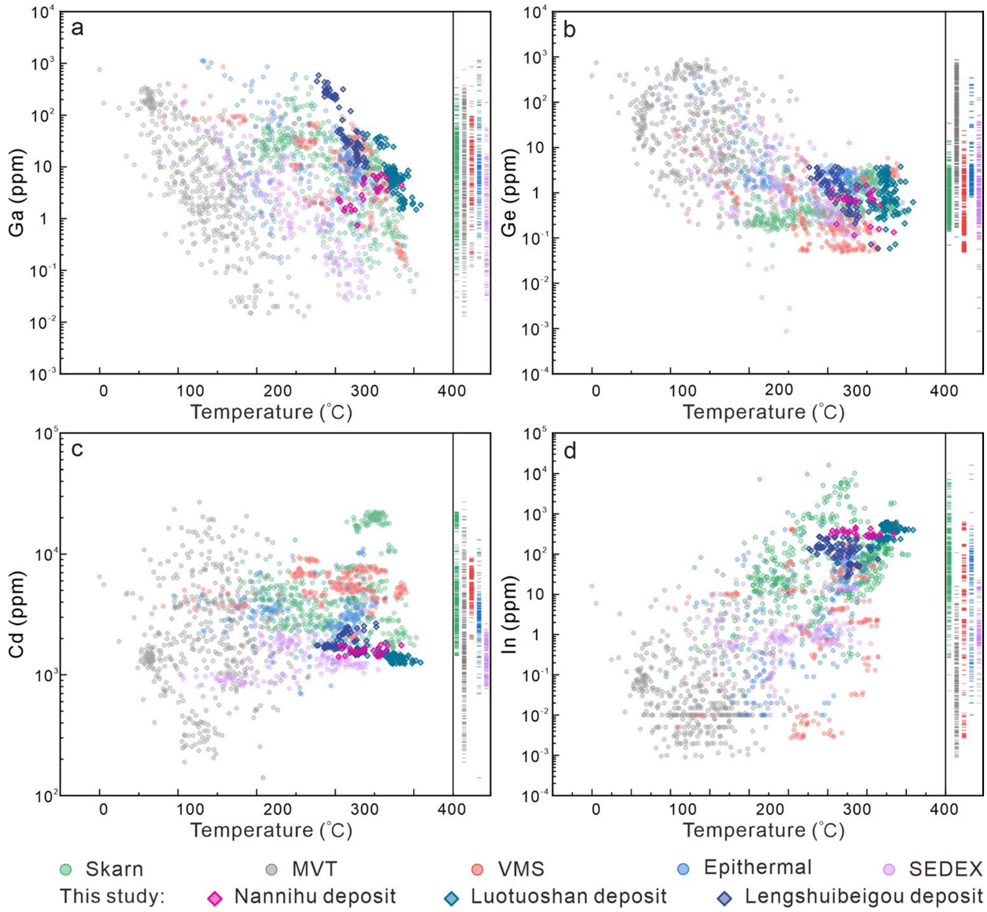


Fig. 11 Correlation plots of (**a**) Ga content vs. calculated sphalerite temperature, (**b**) Ge content vs. Temperature, (**c**) Cd content vs. Temperature, (**d**) In content vs. Temperature. Data comes from different deposit types, listed in ESM 1-Table S4. Rug plots display the distribution of element concentrations in sphalerite from different types of deposits. The temperatures of all deposits are calculated using GGIMFis geothermometer ($T=-54.4*\ln\left(\frac{C_{Ga}^{0.22}C_{Ge}^{0.22}}{C_{Fe}^{0.37}C_{Mn}^{0.20}C_{In}^{0.11}}\right)+208$) (Frenzel et al. 2016)

compared to sphalerite from the epithermal Lengshuibeigou deposit (Figs. 2 and 10d). Gallium shows the opposite trend (Figs. 2 and 10c).

To understand the relationship between Ga and In enrichment or depletion with distance to the causative intrusion in magmatic-hydrothermal systems, the data from the Nannihu ore field is plotted together with some reference data from other typical ore fields or deposits (Fig. 12; ESM 1-Table S5). Each of these deposits has a clear magmatic-hydrothermal origin, a distinct mineralization zonation, and available dispersed element data from sphalerite analyses. In the Nannihu ore field, Ga content increases strongly from the Nannihu and Luotuoshan deposits to the epithermal Lengshuibeigou deposit (Fig. 12a). A similar increase is observed in the Chitudian ore field 20 km SE of the Nannihu ore field, and Baoshan deposits: Ga-enriched sphalerite occurs in distal samples (Fig. 12a; Chen et al. 2023a; Wu et al. 2023). In the Chitudian ore field, samples with high Ga content were collected from distal hydrothermal vein-type orebodies from the Xigou deposit instead of skarn-type orebodies from the Zhongyuku deposit (Fig.2a; Chen et al. 2023a). In the Baoshan deposit, sample BM-16 with high Ga and relatively lower In contents was found in the distal North District, where the contribution of magmatic fluids is minor, and the basement-derived fluid dominates the hydrothermal system (Wu et al. 2023). These datasets confirm previous observations that Ga tends to be enriched in sphalerite from distal epithermal mineralization in the magmatic-hydrothermal systems (Fig. 12a).

Generally, In tends to be enriched in sphalerite from skarn mineralization proximal to the causative intrusion. For example, the highest In concentrations were measured in sphalerite from the Luotuoshan skarn deposit, instead of the Nannihu porphyry deposit (Fig. 12b). The log (Ga/In) values of sphalerite from the Nannihu ore field and Chitudian ore field about 20 km SE of Nannihu, show a positive correlation trend with the distance to ore-forming intrusion, with a linear fit between the average log (Ga/In) value and distance: log (Ga/In) = 0.00037 × distance −1.9793 ($R^2=0.77$; Fig. 12c). Sphalerite analyses from other ore fields and different Pb-Zn districts of China show a similar tendency but have a different slope; the log (Ga/In) values increase from the proximal to distal deposits (Fig. 12d).

The Nannihu ore field and the neighboring Chitudian ore field are both related to late Mesozoic granites in the Luanchuan district and share similar compositions of ore-forming materials (Bao et al. 2009; Tang et al. 2014; Cao et al. 2016; Chen et al. 2023a; Li et al. 2023b). This could explain

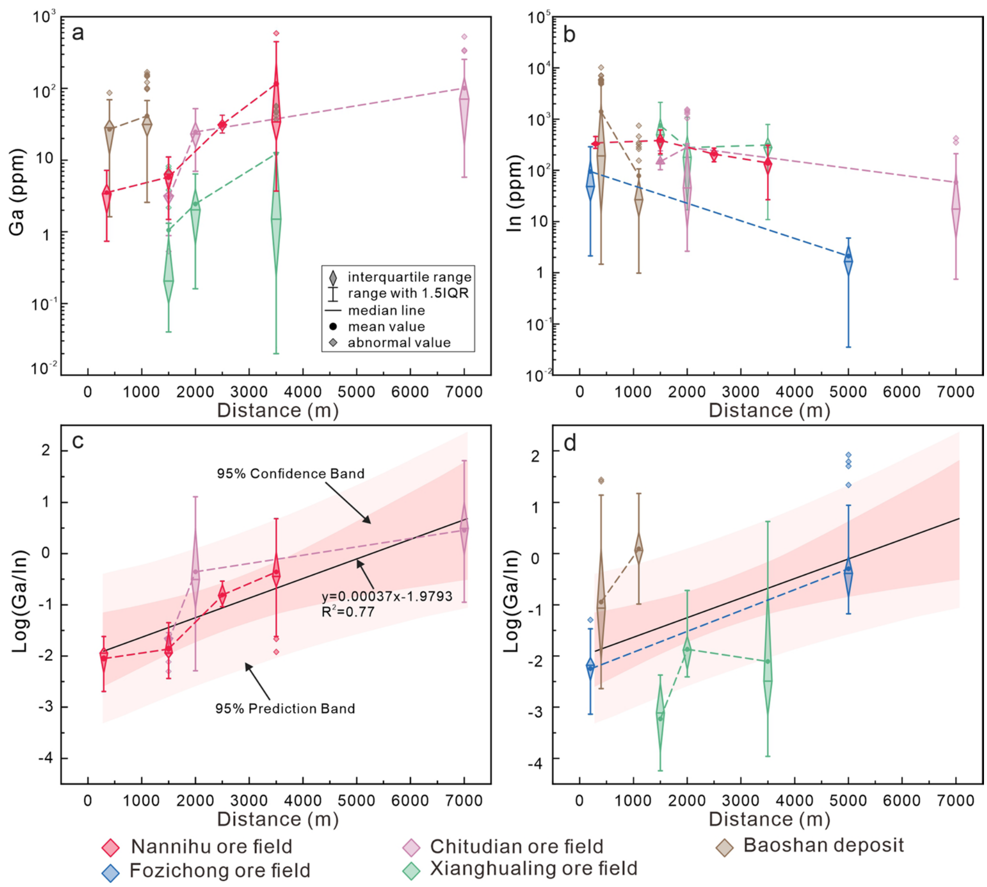


Fig. 12 (**a-b**) Charts showing the spatial distribution of gallium and indium contents. Data from the Nannihu ore field are compared with those from the Chitudian, Fozichong and Xianghualing ore field relative to source intrusions (Tian 2013; Lai 2014; Jia 2022; Chen et al. 2023a; Ma 2023; Tan et al. 2024). Distance refers to distance between the orebody and central ore-forming intrusion, listed in ESM 1-Table S5. (**c**) Correlation plot of log (Ga/In) vs. distance and linear fit of log (Ga/In) average values in Zn-Pb deposits from the Luanchuan district. (**d**) Correlation plot of log (Ga/In) vs. distance to source intrusion in several Zn-Pb deposits from China

the good fit of the result obtained from these deposits. The variation between the slopes of log (Ga/In) values and the distance to the causative intrusion among different ore fields or deposits can be attributed to differences in the metallogenic setting and source material of magmatic-hydrothermal systems (Fig. 12d). Therefore, a common linear model can be formulated to determine the distance of deposits or ore bodies with similar source materials in a same metallogenic province, from the source intrusions.

In a magmatic-hydrothermal system with In and Ga mineralization potential, In exploration should focus on the proximal skarn mineralization within a 2000 m radius of the ore-forming intrusion (Fig. 12b). For example, at the Nannihu deposit, the enrichment of In in sphalerite decreases from the Luotuoshan deposit to the Lengshuibeigou deposit. The In concentration is negatively correlated to the distance from the ore-forming intrusion when the distance extends beyond the range of the skarn mineralization area for In. Based on the spatial distribution of Ga, the distal epithermal mineralization zone in the magmatic-hydrothermal systems is the favorable prospecting area. Given the consistent trend of the log (Ga/In) values of sphalerite from different deposits/ore fields, this could be a useful indicator of the distance between Pb-Zn deposits and the ore-forming intrusions in magmatic-hydrothermal systems, thus contributing to the prediction of the ore-forming center and Pb-Zn exploration targeting.

Conclusions

(1) Indium and Ga enter the sphalerite lattice as the trivalent ions Ga^{3+} and In^{3+}, coupled with available monovalent ions through substitution $Ga^{3+}/In^{3+}+Cu^{+}/Ag^{+}\leftrightarrow 2Zn^{2+}$.
(2) It is essential to select the appropriate geothermometer based on the characteristics of sphalerite when calculating temperature by sphalerite geothermometry. Selecting sphalerite paragenetically associated with other sulfide

minerals such as chalcopyrite or pyrrhotite, improves the accuracy of the GGIMFis geothermometer.

(3) In the magmatic-hydrothermal system, Ga tends to be enriched in the lower-temperature epithermal mineralization, which usually occurs at distal and shallow positions. In contrast, In-enriched sphalerite is predominantly associated with the high-temperature and proximal skarn mineralization and is expected at distances of 0–2000 m from the ore-forming intrusion. The log (Ga/In) of sphalerite can be used to predict the distance from ore bodies to ore-forming intrusions.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00126-025-01392-6.

Acknowledgements We appreciate Mr. Yanyan Sun from the Luanchuan Longyu Molybdenum Industry Co., LTD, and Mr. Guosu Zhang from the Zhongxin Mining Co., LTD for their assistance with the field investigation. We thank Mr. Yizhe Fan from China University of Geosciences (Wuhan) for supporting the laboratory work of fluid-inclusion petrography and microthermometry. Special thanks go to Prof. Carl Spandler for his insightful suggestions during the revision process. We sincerely thank chief editor Prof. Bernd Lehmann, associate editor Prof. Degao Zhai, and two reviewers (Alexandre Cugerone and one anonymous) for their constructive comments that helped to improve the manuscript.

Author contributions CW conducted the analytical experiments and wrote the draft of the manuscript. ZKL supervised this study, provided the conceptualization and revised the manuscript. IS and YZ contributed to the data interpretations and reviewed the manuscript critically. XML and RZZ assisted with fieldwork and data investigation. JH and JZ participated in the LA-ICP-MS analysis of sphalerite and microthermometric measurement of fluid inclusion. All authors contributed to the final version of this manuscript.

Funding This research was jointly supported by the Ministry of Science and Technology of China (2021YFC2900300), National Natural Science Foundation of China (41772074, 42172103), Hubei Provincial Natural Science Foundation of China (2025AFA032), and Geological Exploration Found Program of Henan Province (2024−402).

Data availability The authors confirm that the data supporting the findings of this study are available within the online supplementary materials.

Declarations

Competing interests The authors declare no competing interests.

References

Bao ZW, Zeng QS, Zhao TP, Yuan ZL (2009) Geochemistry and petrogenesis of the ore-related Nannihu and Shangfanggou granite porphyries from East Qinling belt and their constraints on the molybdenum mineralization. Acta Petrol Sin 25:2523–2536 (in Chinese with English abstract)

Bauer ME, Seifert T, Burisch M, Krause J, Richter N, Gutzmer J (2017) Indium-bearing sulfides from the Hämmerlein Skarn deposit, erzgebirge, germany: evidence for late-stage diffusion of indium into sphalerite. Miner Deposita 54:175–192

Bauer ME, Burisch M, Ostendorf J, Krause J, Frenzel M, Seifert T, Gutzmer J (2018) Trace element geochemistry of sphalerite in contrasting hydrothermal fluid systems of the Freiberg district, germany: insights from LA-ICP-MS analysis, near-infrared light microthermometry of sphalerite-hosted fluid inclusions, and sulfur isotope geochemistry. Miner Deposita 54:237–262

Belissont R, Boiron M, Luais B, Cathelineau M (2014) LA-ICP-MS analyses of minor and trace elements and bulk ge isotopes in zoned ge-rich sphalerites from the Noailhac–Saint-Salvy deposit (France): insights into incorporation mechanisms and ore deposition processes. Geochim Cosmochim Acta 126:518–540

Belissont R, Muñoz M, Boiron M, Luais B, Mathon O (2016) Distribution and oxidation state of ge, Cu and Fe in sphalerite by μ-XRF and K-edge μ-XANES: insights into ge incorporation, partitioning and isotopic fractionation. Geochim Cosmochim Acta 177:298–314

Cao HW, Zhang ST, Zheng L, Liu RP, Tian HH, Zhang XH, Li JJ (2014) Geochemical characteristics of trace elements of sphalerite in the Zhongyuku (Pb)-Zn deposit of the luanchuan, Southwest of China. J Mineral Petrol 34:50–59

Cao HW, Pei QM, Zhang ST, Zhang LK, Zheng L, Hu XK (2016) Rb-Sr dating and geological significance of sphalerites from the Zhongyuku Zn(-Pb) deposit in luanchuan, West henan, China. J Chengdu Univ Technol 43:11

Casanova V, Kouzmanov K, Audétat A, Wälle M, Ubrig N, Ortelli M, Fontboté L (2018) Fluid inclusion studies in opaque ore minerals: II. A comparative study of syngenetic synthetic fluid inclusions hosted in quartz and opaque minerals. Econ Geol 113:1861–1883

Chen YJ, Fu SG (1992) Gold mineralization in West henan, China. China Seismological, Beijing. (in Chinese with English abstract)

Chen C, Meng L, Xu J, Zhan Q, Zhao TP (2023a) Texture and geochemistry of sphalerite from the Chitudian Pb-Zn-Ag deposit, Southern margin of the North China craton: implications for the enrichments of cd, ga, and in. Ore Geol Rev 156:105392

Cook NJ, Ciobanu CL, Pring A, Skinner W, Shimizu M, Danyushevsky L, Saini-Eidukat B, Melcher F (2009) Trace and minor elements in sphalerite: A LA-ICPMS study. Geochim Cosmochim Acta 73:4761–4791

Cugerone A, Cenki-Tok B, Muñoz M, Kouzmanov K, Oliot E, Motto-Ros V, Le Goff E (2020) Behavior of critical metals in metamorphosed Pb-Zn ore deposits: example from the pyrenean axial zone. Miner Deposita 56:685–705

Duan SG, Xue CJ, Liu GY, Yan CH, Feng QW, Song YW, Tu QJ, Gao YB, Gao BY (2010) Geology and sulfur isotope geochemistry of lead-zinc deposits in Luanchuan district, Henan province, China. Earth Sci Front 17:375–384 (in Chinese with English abstract)

Fougerouse D, Cugerone A, Reddy SM, Luo K, Motto-Ros V (2023) Nanoscale distribution of ge in Cu-rich sphalerite. Geochim Cosmochim Acta 346:223–230

Frenzel M, Hirsch T, Gutzmer J (2016) Gallium, germanium, indium, and other trace and minor elements in sphalerite as a function of deposit type — A meta-analysis. Ore Geol Rev 76:52–78

Gao XY, Zhao TP, Yuan ZL, Zhou YY, Gao JF (2010) Geochemistry and petrogenesis of the Heyu batholith in the Southern margin of the North China block. Acta Petrol Sin 26:3485–3506

Gao XY, Zhao TP, Bao ZW, Yang AY (2014) Petrogenesis of the early cretaceous intermediate and felsic intrusions at the Southern margin of the North China craton: implications for crust–mantle interaction. Lithos 206–207:65–78

George LL, Cook NJ, Cristiana LC, Wade BP (2015) Trace and minor elements in galena: A reconnaissance LA-ICP-MS study. Am Mineral 100:548–569

George LL, Cook NJ, Ciobanu CL (2016) Partitioning of trace elements in co-crystallized sphalerite–galena–chalcopyrite hydrothermal ores. Ore Geol Rev 77:97–116

Graupner T, Henning S, Goldmann S, Fuchs S, Stedingk K, Liessmann W, Birkenfeld S (2024) The In-Ga-Sb association of the post-Variscan Zn-Pb-Ag vein deposit at lautenthal, upper Harz mountains, germany: sphalerite mineral chemistry. Miner Deposita 59:1363–1386

Gunn G (2014) Critical metals handbook, 1st edn. John Wiley & Sons, Hoboken, NJ

Guo BJ (2006) Study on the Mesozoic metallogenic associations and regularities of the east Qinling, China. Ph. D. thesis. China University of Geosciences (Beijing) (in Chinese with English abstract)

Han YG, Zhang SH, Pirajno F, Zhang YH (2011) Evolution of the mesozoic granites in the Xiong'ershan-Waifangshan region, Western Henan province, china, and its tectonic implications. Acta Geol Sinica - Engl Ed 81:253–265

Hu SX, Lin QL, Chen ZM, Li SM (1988) Geology and metallogeny of the collision belt between the North and the South China plates. Nanjing University Press, Nanjing 1-558 (in Chinese with English abstract)

Jia R (2022) GOCAD 3D geological modeling and deep metallogenic prediction of the Chitudian Xigou Deposit, Henan Province. Ph. D. thesis. China University of Geosciences (Beijing) (in Chinese with English abstract)

Johan Z (1988) Indium and germanium in the structure of sphalerite: an example of coupled substitution with copper. Min Petrol 39:211–229

Keith M, Haase KM, Schwarz-Schampera U, Klemd R, Petersen S, Bach W (2014) Effects of temperature, sulfur, and oxygen fugacity on the composition of sphalerite from submarine hydrothermal vents. Geology 42:699–702

Lai SH (2014) Research on mineralization of the Xianghualing tin polymetallic deposit, Hunan Province, China. Ph. D. thesis. China University of Geosciences (Beijing) (in Chinese with English abstract)

Li YF (2005) The temporal-spital evolution of Mesozoic granitoids in the Xiong'ershan Area and their relationship to molybdenum-gold mineralization. Ph. D. thesis. China University of Geosciences (Beijing) (in Chinese with English abstract)

Li ZK (2013) Metallogenesis of the silver-lead-zinc deposits along the southern margin of the North China Craton. Ph. D. thesis. China University of Geosciences (Wuhan) (in Chinese with English abstract)

Li JW, Bi SJ, Selby D, Chen L, Vasconcelos P, Thiede D, Zhou MF, Zhao XF, Li ZK, Qiu HN (2012a) Giant mesozoic gold provinces related to the destruction of the North China craton. Earth Planet Sci Lett 349–350:26–37

Li JW, Li ZK, Zhou MF, Chen L, Bi SJ, Deng XD, Qiu HN, Cohen B, Selby D, Zhao XF (2012b) The early cretaceous Yangzhaiyu lode gold deposit, North China craton: A link between craton reactivation and gold veining. Econ Geol 107:43–79

Li ZK, Li JW, Zhao XF, Zhou MF, Selby D, Bi SJ, Sui JX, Zhao ZJ (2013) Crustal-extension Ag-Pb-Zn veins in the xiong'ershan district, Southern North China craton: constraints from the Shagou deposit. Econ Geol 108:1703–1729

Li ZK, Li JW, Cooke DR, Danyushevsky L, Zhang L, O'Brien H, Lahaye Y, Zhang W, Xu HJ (2016) Textures, trace elements, and Pb isotopes of sulfides from the Haopinggou vein deposit, Southern North China craton: implications for discrete Au and Ag-Pb-Zn mineralization. Contrib Mineral Petrol 171:99

Li ZK, Bi SJ, Li JW, Zhang W, Cooke DR, Selby D (2017) Distal Pb-Zn-Ag veins associated with the world-class Donggou porphyry mo deposit, Southern North China craton. Ore Geol Rev 82:232–251

Li XM, Zhang YX, Li ZK, Zhao X, Zuo R, Xiao F, Zheng Y (2023a) Discrimination of Pb-Zn deposit types using sphalerite geochemistry: new insights from machine learning algorithm. Geosci Front 14:101580

Li ZK, Wang C, Wang ZQ, Li KW, Zhou JJ, Li XM, Hu J, Zhang J (2023b) A giant porphyry-skarn-epithermal Mo-W-Cu-Zn-Pb-Ag metallogenic system in the Nannihu ore field, Southern North China craton: constraints from new geochronological and pyrite geochemical data. Ore Geol Rev 163:105792

Liu YS, Hu ZC, Gao S, Günther D, Xu J, Gao CG, Chen HH (2008) In situ analysis of major and trace elements of anhydrous minerals by LA-ICP-MS without applying an internal standard. Chem Geol 257:34–43

Liu TG, Ye L, Zhou JX, Wang XL (2010a) Cd primarily isomorphously replaces Fe but not Zn in sphalerite. Acta Mineral Sinica 30:179–184 (in Chinese with English abstract)

Liu YS, Hu ZC, Zong KQ, Gao CG, Gao S, Xu J, Chen HH (2010b) Reappraisement and refinement of Zircon U-Pb isotope and trace element analyses by LA-ICP-MS. Chin Sci Bull 55:1535–1546

Liu WH, Mei Y, Etschmann B, Glenn M, MacRae CM, Spinks SC, Ryan CG, Brugger J, Paterson DJ (2023) Germanium speciation in experimental and natural sphalerite: implications for critical metal enrichment in hydrothermal Zn-Pb ores. Geochim Cosmochim Acta 342:198–214

Luo K, Cugerone A, Zhou MF, Zhou JX, Sun GT, Xu J, He KJ, Lu MD (2022) Germanium enrichment in sphalerite with acicular and euhedral textures: an example from the Zhulingou carbonate-hosted Zn(-Ge) deposit, South China. Miner Deposita 57:1343–1365

Ma XH (2023) Study on the Indium occurrence state and enrichment regularity in Xianghualing Sn-Pb-Zn Orefield, Southern Hunan. Master's thesis. East China University of Technology, Jiangxi, (in Chinese with English abstract)

Mao JW, Xie GQ, Bierlein F, Qu WJ, Du AD, Ye HS, Pirajno F, Li HM, Guo BJ, Li YF (2008) Tectonic implications from Re-Os dating of mesozoic molybdenum deposits in the East Qinling-Dabie orogenic belt. Geochim Cosmochim Acta 72:4607–4626

Mao JW, Xie GQ, Pirajno F, Ye HS, Wang YB, Li YF, Xiang JF, Zhao HJ (2010) Late Jurassic-Early cretaceous granitoid magmatism in Eastern qinling, central-eastern china: SHRIMP Zircon U-Pb ages and tectonic implications. Aust J Earth Sci 57:51–78

Mao JW, Pirajno F, Xiang JF, Gao JJ, Ye HS, Li YF, Guo BJ (2011) Mesozoic molybdenum deposits in the East Qinling-Dabie orogenic belt: characteristics and tectonic settings. Ore Geol Rev 43:264–293

Moritz R (2006) Fluid salinities obtained by infrared microthermometry of opaque minerals: implications for ore deposit modeling — A note of caution. J Geochem Explor 89:284–287

Murakami H, Ishihara S (2013) Trace elements of Indium-bearing sphalerite from tin-polymetallic deposits in bolivia, China and japan: A femto-second LA-ICPMS study. Ore Geol Rev 53:223–243

Niu PP, Muñoz M, Mathon O, Xiong SF, Jiang SY (2024) Mechanism of germanium enrichment in the world-class Huize MVT Pb–Zn deposit, Southwestern China. Miner Deposita 59:995–1016

Ohta E (1995) Common features and genesis of tin-polymetallic veins. Resour Geol 18:187–195

Peng P, Zhai M, Ernst RE, Guo J, Liu F, Hu B (2008) A 1.78 Ga large igneous Province in the North China craton: the xiong'er volcanic Province and the North China Dyke swarm. Lithos 101:260–280

Peng HW, Fan HR, Santosh M, Hu FF, Jiang P (2020) Infrared microthermometry of fluid inclusions in transparent to opaque minerals: challenges and new insights. Miner Deposita 55:1425–1440

Schwartz MO (2002) Cadmium in zinc deposits: economic geology of a polluting element. Int Geol Rev 42:445–469

Steele-MacInnis M, Lecumberri-Sanchez P, Bodnar RJ (2012) HokieFlincs_H_2O-NaCl: A Microsoft excel spreadsheet for interpreting microthermometric data from fluid inclusions based on the PVTX properties of H_2O–NaCl. Comp Geosci 49:334–337

Tan ZJ, Zheng Y, Yu PP, Li RJ, Huang Y, Ding W, Wu YH, Chen MX, Sun LH, Wang ZK, Chen X (2024) Indium concentrations as a potential indicator of orebody-intrusion distance in a Skarn Pb-Zn system, elucidated by the Fozichong orefield (South China). Ore Geol Rev 165:105912

Tang L, Zhang ST, Cao HW, Li D, Zhang XH, Zhang YH, Tian HH, Zhang HH (2014) Metallogenic system and evolutionary characteristics of Mo-W-Pb-Zn-Ag polymetallic metallogenic concentration area in luanchuan, henan, China. J Chengdu Univ Technol (Science Technol Edition) 41(3):356–368 (in Chinese with English abstract)

Tian HH (2013) Geological, geochemical characteristics and genesis of the Chitudian lead-zinc deposit, West Henan. Master's thesis. China University of Geosciences (Beijing) (in Chinese with English abstract)

Torró L, Millán-Nuñez AJ, Benites D, González-Jiménez JM, Laurent O, Tavazzani L, Vallance J, Chelle-Michou C, Proenza JA, Flores C, Melgarejo JC, Rosas S, Fontboté L (2023) Germanium- and gallium-rich sphalerite in Mississippi Valley–type deposits: the San Vicente district and the Shalipayco deposit, Peru. Miner Deposita 58:853–880

Tu GC, Gao ZM, Hu RZ (2004) The geochemistry and ore-forming mechanism of the dispersed elements. Geology, Beijing. (in Chinese)

Wang CM, Chen L, Bagas L, Lu YJ, He XY, Lai XR (2015) Characterization and origin of the Taishanmiao aluminous A-type granites: implications for early cretaceous lithospheric thinning at the Southern margin of the North China craton. Int J Earth Sci 105:1563–1589

Wu F, Lin J, Wilde SA, Zhang X, Yang J (2005) Nature and significance of the early cretaceous giant igneous event in Eastern China. Earth Planet Sci Lett 233:103–119

Wu YH, Yu PP, Chen X, Hu ZB, Long LJ, He B, Zheng Y (2023) Earlier stage, higher temperature, and deeper space facilitate indium precipitation in a Skarn system, as exemplified by the Baoshan Pb-Zn polymetallic deposit, South China. Ore Geol Rev 163:105745

Xiang JF, Mao JW, Pei RF, Ye HS, Wang CY, Tian ZH, Wang HL (2012) New geochronological data of granites and ores from the Nannihu-Sandaozhuang mo (W) deposit. Geol China 39:458–473 (in Chinese with English abstract)

Xiao E, Hu J, Zhang ZZ, Dai BZ, Wang YF, Li HY (2012) Petrogeochemistry, Zircon U-Pb dating and Lu-Hf isotopic compositions of the Haoping and Jinshanmiao granites from the Huashan complex batholith in Eastern Qinling orogen. Acta Petrol Sin 28(12):4031–4046 (in Chinese with English abstract)

Xu J, Cook NJ, Ciobanu CL, Li XF, Kontonikas-Charos A, Gilbert S, Lv Y (2020) Indium distribution in sphalerite from sulfide–oxide–silicate Skarn assemblages: a case study of the Dulong Zn–Sn–In deposit, Southwest China. Miner Deposita 56:307–324

Xu J, Ciobanu CL, Cook NJ, Slattery A, Li XF, Kontonikas-Charos A (2021) Phase relationships in the system ZnS-$CuInS_2$: insights from a nanoscale study of indium-bearing sphalerite. Am Mineral 106:192–205

Yan CH (2004) Study on inner structure of Pb-Zn-Ag mineralization system in Eastern Qingling. Geology Publishing House, Beijing, p 144. (in Chinese with English abstract)

Yan CH, Liu GY, Peng Y, Song YW, Wang JZ, Zhao RJ, Zeng XY, Lv WD, Yao XN, Ma HW (2009) The metallogenetical characteristics of the lead-zinc-silver deposits in Southwest Henan. Geological, Beijing, p 365. (in Chinese)

Ye HS (2006) The Mesozoic tectonic evolution and Pb-Zn-Ag metallogeny in the south margin of North China Craton. Ph. D. thesis. Chinese Academy of Geological Sciences, Beijing (in Chinese with English abstract)

Ye HS, Mao JW, Li YF, Yan CH, Guo BJ, Zhao C, He C, Zheng R, Chen L (2006) Characteristics and metallogenic mechanism of Mo-W and Pb-Zn-Ag deposits in Nannihu ore field, Western Henan Province. Geoscience 20(1):165–174 (in Chinese with English abstract)

Ye HS, Mao JW, Xu LG, Gao JJ, He CF (2008) SHRIMP Zircon U-Pb dating and geochemistry of the Taishanmiao aluminous A-type granite in Western Henan Province. Geol Rev 54:699–711

Ye L, Cook NJ, Ciobanu CL, Liu YPL, Zhang Q, Liu TG, Wei G, Danyushevskiy YYL L (2011) Trace and minor elements in sphalerite from base metal deposits in South china: A LA-ICPMS study. Ore Geol Rev 39:188–217

Yun H, Guo B, Hu HL, Zhu HY, Han JW, Yan HQ, Li W (2020) Geological characteristics and genesis of the Luotuoshan copper-zinc polymetallic deposit in Luanchuan county. Metal Mine 11:152–161

Zhang JK, Shao YJ, Liu ZF, Chen K (2022) Sphalerite as a record of metallogenic information using multivariate statistical analysis: constraints from trace element geochemistry. J Geochem Explor 232:106883

Zhang RZ, Bao B, Guo B, Han JW, Li KW, Yun H, Zhu HY, Li M, Sun YY (2023) Distribution and enrichment characteristics of indium in the Luotuoshan sulfur polymetallic deposit,henan Province. Acta Geol Sinica 97:2547–2562

Zhao TP, Zhou MF, Zhai MG, Xia B (2002) Paleoproterozoic rift-related volcanism of the xiong'er group, North China craton: implications for the breakup of Columbia. Int Geol Rev 44:336–351

Zhao TP, Meng L, Gao XY, Jin C, Wu Q, Bao ZW (2018) Late mesozoic felsic magmatism and Mo-Au-Pb-Zn mineralization in the Southern margin of the North China craton: A review. J Asian Earth Sci 161:103–121

Zhao XF, Li ZK, Zhao SR, Bi SJ, Li JW (2019) Early cretaceous regional-scale magmatic-hydrothermal metallogenic system at the Southern margin of the North China craton. Earth Sci 44:52–68 (in Chinese with English abstract)

Zhao HT, Zhang Y, Shao YJ, Liao J, Song SL, Cao GS, Tan RC (2024) A new sphalerite thermometer based on machine learning with trace element geochemistry. Nat Resour Res 33:2609–2626

Zhu CW, Wen HJ, Zhang YX, Fu SH, Fan HF, Cloquet C (2016) Cadmium isotope fractionation in the Fule Mississippi Valley-type deposit, Southwest China. Miner Deposita 52: 675-686.

Zhu RX (2011) Timing, scale and mechanism of the destruction of the North China Craton. Sci China Ser D 41(5): 583-592 (in Chinese)